DUALITY IN DECONFOUNDING WITH INSTRUMENTS AND SURROGATE CONTROLS

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Abstract: Instrumental variables and surrogate controls address unmeasured confounding through different sources of variation, but either source alone may fail to identify all structural coefficients or may be invalid. We develop a projected-moment framework for linear structural models whose moment conditions are based respectively on the component of the regressors explained by instruments and the component orthogonal to surrogate controls. We characterize the statistical target of each source under moment violations, accounting for estimated projections, and show that the combined moment conditions can identify the full coefficient vector even when neither source does alone. The jointly identified coefficient is estimated by generalized method of moments, with weights chosen to minimize worst-case first-order risk over a prespecified set of local moment violations. Under exact specification, efficient projected GMM has no larger trace of asymptotic variance than identified single-source estimators. Under local misspecification, risk-optimal weighting achieves no larger worst-case first-order risk over the specified sensitivity set. Simulations and an application to single-cell

perturbation data illustrate the efficiency gains and the sensitivity of the findings to local moment violations.

Key words and phrases: Generalized method of moments, instrumental variables, local misspecification, surrogate controls.

1. Introduction

Estimating the causal effect of an exposure on an outcome is challenging in the presence of unmeasured confounders. Experimental perturbations help by inducing variation in the exposure, but they do not remove this difficulty, because the exposure need not be randomized and it is measured after the experiment. This problem arises in Perturb-seq experiments, in which cells receive pooled CRISPR perturbations targeting individual genes and single-cell RNA sequencing then measures the expression of both the targeted gene and candidate outcome genes (Dixit et al., 2016; Replogle et al., 2022; Du et al., 2026). We seek to estimate the causal effect of target-gene expression on outcome-gene expression, in the presence of unmeasured confounders, including unobserved cellular states and technical factors that may influence both measurements (Du et al., 2025).

These experiments provide two sources of auxiliary information for addressing this confounding. The first is perturbation assignment, which may

serve as an instrumental variable (IV) by providing experimentally induced variation in target-gene expression (Wu and Zhou, 2026). Classical IV analysis requires the assignment to be relevant for target-gene expression, independent of the unmeasured confounders, and to affect outcome-gene expression only through the target gene. The second is additional measurements of cell states, which may serve as surrogate controls (SV) if they capture the information needed to adjust for unmeasured confounding (Du et al., 2026). Either source may fail to identify every structural coefficient, and violations of its validity conditions can introduce bias. When both are available, the questions are which coefficients they jointly identify, how precisely, and how sensitive the resulting estimates are to violations of these conditions.

Classical IV methods and generalized method of moments (GMM) use instrumental moment conditions for identification and estimation (Sargan, 1958; Hansen, 1982). Proxy-variable and proximal methods address unmeasured confounding using auxiliary measurements; proximal identification combines treatment-inducing and outcome-inducing proxies through a confounding bridge under suitable assumptions (Miao et al., 2018; Tchetgen Tchetgen et al., 2024). Related work studies identification using auxiliary variables for multiple treatments (Miao et al., 2023) and negative controls with invalid instruments (Dukes et al., 2025). Sensitivity analysis

under approximate moment conditions addresses uncertainty about moment validity (Armstrong and Kolesár, 2021). Our setting considers instruments and surrogate controls that provide separate moment conditions for a common linear structural coefficient. This raises questions about when these conditions jointly identify the coefficient, even if neither suffices alone, and how their information should be combined when validity is uncertain.

To address these questions, we construct two sets of moment conditions for a linear structural model by projecting the exposures. The IV moments use the component of the exposures explained by the instruments. The SV moments use the component remaining after linear adjustment for the surrogate controls. Under their respective validity conditions, both sets hold at the same structural coefficient, and their common projected form allows them to be stacked for joint estimation.

Building on this formulation, we characterize the statistical targets, identification conditions, and asymptotic distributions of the IV and SV estimators, accounting for estimated projections. For the combined system, we establish a necessary and sufficient rank condition for identification, including cases where neither source alone identifies the full coefficient vector, and develop GMM inference under exact specification and local misspecification. To address uncertainty about moment validity, we derive weights

minimizing worst-case first-order risk over a prespecified ellipsoidal set of local moment violations and construct a feasible estimator with the same first-order limit as the oracle. Under the respective assumptions, efficient combined GMM has no larger trace of asymptotic variance than identified single-source estimators, while risk-optimal weighting has no larger worst-case first-order risk. Simulation studies and a CRISPR perturbation application examine finite-sample performance and sensitivity to moment violations. Sections 2–4 develop the methodology, Sections 5–6 present numerical and empirical analyses, and Section 7 concludes.

2. Projected moments for IV and SV

2.1 Structural model and auxiliary variables

Throughout the population analysis, we work with centered variables: $\mathbb{E}(x) = 0$, $\mathbb{E}(z) = 0$, and $\mathbb{E}(w) = 0$. The intercept is therefore omitted from the model.

Consider the confounded linear regression model

$$y = x^\top \beta + \varsigma, \tag{2.1}$$

where $y \in \mathbb{R}$, $x \in \mathbb{R}^p$, $\beta \in \mathbb{R}^p$, and $\varsigma \in \mathbb{R}$. We assume $\mathbb{E}(\varsigma) = 0$, $\mathrm{Var}(\varsigma) = \sigma^2 < \infty$, and $\Sigma_x := \mathbb{E}(xx^\top) \in \mathbb{S}^p_{++}$. When $\mathbb{E}(x\varsigma) = 0$, or-

dinary least squares consistently estimates β. Our interest is instead in settings where $\mathbb{E}(x\varsigma) \neq 0$, for example because the disturbance contains a latent confounding component $u \in \mathbb{R}^r$ associated with both the regressors and the outcome. An illustrative causal structure is shown in Figure 1, but our statistical analysis does not rely on the directions of the edges between u and w or between z and x.

For n independent and identically distributed (i.i.d.) observations, write the sample model as

$$Y = X\beta + \varsigma, \tag{2.2}$$

where $Y = (y_1, \ldots, y_n)^\top \in \mathbb{R}^n$, $X = (x_1, \ldots, x_n)^\top \in \mathbb{R}^{n \times p}$, and $\varsigma = (\varsigma_1, \ldots, \varsigma_n)^\top \in \mathbb{R}^n$.

To address latent confounding, we use observed auxiliary variables that provide information beyond x_i. The IV and SV channels exploit, respectively, instrument-induced and residualized variation in x_i. Their distinct projection geometries are developed below.

2.2 Instrumental variable models

Suppose an instrument vector $z_i \in \mathbb{R}^q$ is observed for each i. Under the centered formulation, let $\Sigma_z := \mathbb{E}(z_i z_i^\top)$, $\Sigma_{xz} := \mathbb{E}(x_i z_i^\top)$, and $\Sigma_{zx} := \Sigma_{xz}^\top$. When $\Sigma_z \in \mathbb{S}_{++}^q$, the population linear projection is $\Pi_z x_i := \Sigma_{xz}\Sigma_z^{-1} z_i$,

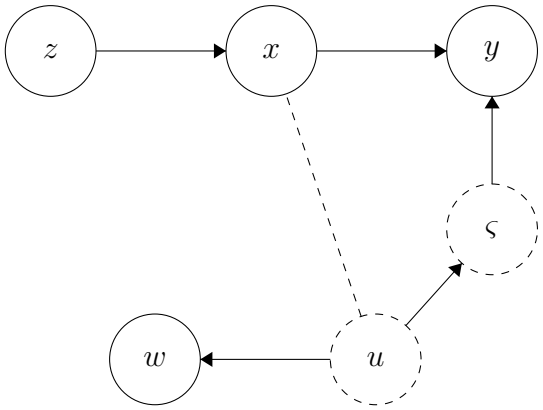


Figure 1: Illustrative causal diagram for instrumental variables and surrogate controls. The diagram imposes stronger causal restrictions than the projected moment conditions used in the analysis. Solid and dashed circles denote observed and unobserved variables, respectively.

which retains the component of x_i linearly explained by the instruments.

The IV approach uses the projected moment equation $\mathbb{E}\{(\Pi_z x_i)(y_i - x_i^\top \theta)\} = 0_p$ for a candidate coefficient θ. Define the relevance matrix $Q_{\text{IV}} := \mathbb{E}\{(\Pi_z x_i)x_i^\top\} = \Sigma_{xz}\Sigma_z^{-1}\Sigma_{zx}$ and projected-moment bias $\delta_{\text{IV}} := \mathbb{E}\{(\Pi_z x_i)\varsigma_i\}$. Substituting the structural model gives $Q_{\text{IV}}(\beta - \theta) + \delta_{\text{IV}} = 0_p$. When $Q_{\text{IV}} \in \mathbb{S}^p_{++}$, its unique solution is the statistical target $\beta_{\text{IV}} := \beta + Q_{\text{IV}}^{-1}\delta_{\text{IV}}$. The raw IV restriction $\mathbb{E}(z_i\varsigma_i) = 0_q$ implies $\delta_{\text{IV}} = 0_p$, so this target equals β. Otherwise, a nonzero δ_{IV} shifts the target away from β.

At the sample level, let $Z = (z_1, \ldots, z_n)^\top \in \mathbb{R}^{n\times q}$ and $P_Z := Z(Z^\top Z)^{-1}Z^\top$.

The first stage projects the regressors onto the instrument span, giving the fitted design $\widehat{X} := P_Z X$. Whenever $Z^\top Z$ and $X^\top P_Z X$ are nonsingular, the second-stage estimator is

$$\widehat{\beta}_{\mathrm{IV}} := \operatorname*{argmin}_{\theta \in \mathbb{R}^p} \|Y - \widehat{X}\theta\|_2^2 = (X^\top P_Z X)^{-1} X^\top P_Z Y. \tag{2.3}$$

Equivalently, the sample rank conditions are $\operatorname{rank}(Z) = q$ and $\operatorname{rank}(Z^\top X) = p$, which necessarily require $p \leq q \leq n$. Throughout the fixed-dimensional asymptotic analysis, p and q are treated as fixed as $n \to \infty$.

Under raw IV validity, $\mathbb{E}\{z_i(y_i - x_i^\top \beta_{\mathrm{IV}})\} = 0_q$, so the raw moment is centered. Projected validity alone does not ensure this condition. To obtain a centered term for the asymptotic expansion, define $e_{\mathrm{IV},i} := y_i - x_i^\top \beta_{\mathrm{IV}} - z_i^\top \Sigma_z^{-1} \mathbb{E}\{z_i(y_i - x_i^\top \beta_{\mathrm{IV}})\}$. Then $\mathbb{E}(z_i e_{\mathrm{IV},i}) = 0_q$. When the raw moment has a nonzero mean, estimation of the first-stage projection contributes at first order, so the usual score $(\Pi_z x_i)(y_i - x_i^\top \beta_{\mathrm{IV}})$ is generally insufficient. Accounting for this contribution gives $\psi_{\mathrm{IV},i} := (\Pi_z x_i) e_{\mathrm{IV},i} + x_i z_i^\top \Sigma_z^{-1} \mathbb{E}\{z_i(y_i - x_i^\top \beta_{\mathrm{IV}})\}$, with covariance $\Sigma_{\mathrm{IV}} := \operatorname{Var}(\psi_{\mathrm{IV},i})$.

Assumption 1 (Projected instrumental variables conditions)**.** Suppose the following conditions hold.

(i) *Projected relevance.* Suppose $\Sigma_z \in \mathbb{S}_{++}^q$ and $Q_{\mathrm{IV}} \in \mathbb{S}_{++}^p$.

(ii) *Second moments and nondegeneracy.* Suppose $\mathbb{E}[\|x_i z_i^\top\|_{\mathrm{op}}^2] < \infty$, $\mathbb{E}\|z_i e_{\mathrm{IV},i}\|_2^2 < \infty$, and $\Sigma_{\mathrm{IV}} \in \mathbb{S}_{++}^p$.

Theorem 1 (Identification and asymptotic properties of projected 2SLS)**.** Under Assumption 1(i), the following statements hold. (i) *Identification.* The projected population moment equation $\mathbb{E}[(\Pi_z x_i)(y_i - x_i^\top \theta)] = 0$ has the unique solution $\theta = \beta_{\mathrm{IV}}$. (ii) *Consistency.* As $n \to \infty$, $\widehat{\beta}_{\mathrm{IV}} \xrightarrow{\mathrm{p}} \beta_{\mathrm{IV}}$. (iii) *Asymptotic normality.* If, in addition, Assumption 1(ii) holds, then $\sqrt{n}(\widehat{\beta}_{\mathrm{IV}} - \beta_{\mathrm{IV}}) \xrightarrow{\mathrm{d}} \mathcal{N}(0, V_{\mathrm{IV}})$, where $V_{\mathrm{IV}} := Q_{\mathrm{IV}}^{-1} \Sigma_{\mathrm{IV}} Q_{\mathrm{IV}}^{-1}$.

The relevance condition in Theorem 1 requires sufficient instrument-induced variation to identify all p coefficients. Since $\mathrm{rank}(Q_{\mathrm{IV}}) = \mathrm{rank}(\Sigma_{zx}) \leq \min(p, q)$, this condition is equivalent to $\mathrm{rank}(\Sigma_{zx}) = p$ and necessarily requires $q \geq p$.

Identification does not, however, imply that projection retains all information in the raw IV moments. When $q > p$, projection may discard overidentifying information; see Appendix B.1 for further discussion.

2.3 Surrogate controls and partialling out

Suppose a surrogate-control vector $w_i \in \mathbb{R}^m$ is observed for each i. Under the centered formulation, let $\Sigma_w := \mathbb{E}(w_i w_i^\top)$ and $\Sigma_{xw} := \mathbb{E}(x_i w_i^\top)$. When $\Sigma_w \in \mathbb{S}_{++}^m$, the population linear projection is $\Pi_w x_i := \Sigma_{xw} \Sigma_w^{-1} w_i$. The SV

approach retains the residual $\Pi_w^{\perp} x_i := x_i - \Pi_w x_i$, which is orthogonal to the surrogate-control span.

The SV approach uses the residualized moment equation $\mathbb{E}\{(\Pi_w^{\perp} x_i)(y_i - x_i^{\top}\theta)\} = 0_p$ for a candidate coefficient θ. Define the relevance matrix $Q_{\mathrm{SV}} := \mathbb{E}\{(\Pi_w^{\perp} x_i) x_i^{\top}\}$ and residualized-moment bias $\delta_{\mathrm{SV}} := \mathbb{E}\{(\Pi_w^{\perp} x_i)\varsigma_i\}$. Substituting the structural model gives $Q_{\mathrm{SV}}(\beta - \theta) + \delta_{\mathrm{SV}} = 0_p$. When $Q_{\mathrm{SV}} \in \mathbb{S}_{++}^p$, its unique solution is the statistical target $\beta_{\mathrm{SV}} := \beta + Q_{\mathrm{SV}}^{-1}\delta_{\mathrm{SV}}$. Exact SV validity requires $\delta_{\mathrm{SV}} = 0_p$, so this target equals β, without requiring $\mathbb{E}(x_i\varsigma_i) = 0_p$ or exogeneity of w_i. A nonzero δ_{SV} represents residual confounding after surrogate adjustment and shifts the target away from β.

At the sample level, let $W = (w_1, \ldots, w_n)^{\top} \in \mathbb{R}^{n\times m}$, $P_W := W(W^{\top}W)^{-1}W^{\top}$, and $P_W^{\perp} := I_n - P_W$. The first step partials out the surrogate controls from both the regressors and the outcome, giving $\widetilde{X} := P_W^{\perp} X$ and $\widetilde{Y} := P_W^{\perp} Y$. When $W^{\top}W$ and $X^{\top}P_W^{\perp}X$ are nonsingular, the second-step estimator is

$$\widehat{\beta}_{\mathrm{SV}} := \operatorname*{argmin}_{\theta\in\mathbb{R}^p} \|\widetilde{Y} - \widetilde{X}\theta\|_2^2 = (X^{\top}P_W^{\perp}X)^{-1}X^{\top}P_W^{\perp}Y. \tag{2.4}$$

Equivalently, the sample rank conditions are $\operatorname{rank}(W) = m$ and $\operatorname{rank}(P_W^{\perp}X) = p$, requiring $m + p \le n$; throughout, m and p are fixed as $n \to \infty$.

Residualized validity does not require $\mathbb{E}\{w_i(y_i - x_i^{\top}\beta_{\mathrm{SV}})\} = 0_m$. To obtain a centered moment for the asymptotic expansion, define $e_{\mathrm{SV},i} := y_i - x_i^{\top}\beta_{\mathrm{SV}} - w_i^{\top}\Sigma_w^{-1}\mathbb{E}\{w_i(y_i - x_i^{\top}\beta_{\mathrm{SV}})\}$. Then $\mathbb{E}(w_i e_{\mathrm{SV},i}) = 0_m$. Unlike the IV

projection, the sample residual-maker satisfies $P_W^{\perp} W = 0$, so the component removed from the residual is eliminated rather than retained. Accounting for estimation of the surrogate projection therefore gives the first-order score $\psi_{\mathrm{sv},i} := (\Pi_w^{\perp} x_i) e_{\mathrm{sv},i}$, with covariance $\Sigma_{\mathrm{sv}} := \mathrm{Var}(\psi_{\mathrm{sv},i})$.

Assumption 2 (Partialling-out conditions with observed controls)**.** Suppose the following conditions hold.

(i) *Residualized relevance.* Suppose $\Sigma_w \in \mathbb{S}_{++}^m$ and $Q_{\mathrm{sv}} \in \mathbb{S}_{++}^p$.

(ii) *Second moments and nondegeneracy.* Suppose $\mathbb{E}\|x_i \varsigma_i\|_2^2 < \infty$, $\mathbb{E}\|x_i x_i^\top\|_{\mathrm{op}}^2 < \infty$, $\mathbb{E}\|x_i w_i^\top\|_{\mathrm{op}}^2 < \infty$, $\mathbb{E}\|w_i e_{\mathrm{sv},i}\|_2^2 < \infty$, and $\Sigma_{\mathrm{sv}} \in \mathbb{S}_{++}^p$.

Theorem 2 (Identification and asymptotic properties of partialling out)**.** Under Assumption 2(i), the following statements hold. (i) *Identification.* The residualized population moment equation $\mathbb{E}[(\Pi_w^{\perp} x_i)(y_i - x_i^\top \theta)] = 0$ has the unique solution $\theta = \beta_{\mathrm{sv}}$. (ii) *Consistency.* As $n \to \infty$, $\widehat{\beta}_{\mathrm{sv}} \xrightarrow{\mathrm{p}} \beta_{\mathrm{sv}}$. (iii) *Asymptotic normality.* If, in addition, Assumption 2(ii) holds, then $\sqrt{n}(\widehat{\beta}_{\mathrm{sv}} - \beta_{\mathrm{sv}}) \xrightarrow{\mathrm{d}} \mathcal{N}(0, V_{\mathrm{sv}})$, where $V_{\mathrm{sv}} := Q_{\mathrm{sv}}^{-1} \Sigma_{\mathrm{sv}} Q_{\mathrm{sv}}^{-1}$.

Under exact surrogate-control validity, $\delta_{\mathrm{sv}} = 0_p$, we have $\beta_{\mathrm{sv}} = \beta$ and hence $\widehat{\beta}_{\mathrm{sv}} \xrightarrow{\mathrm{p}} \beta$. Identification requires sufficiently rich variation in x_i to remain after partialling out the surrogate controls; see Appendix B.2 for further discussion of the residualization geometry.

2.4 Causal interpretation

Theorems 1 and 2 identify the statistical targets $\beta_{\text{IV}} = \beta + Q_{\text{IV}}^{-1}\delta_{\text{IV}}$ and $\beta_{\text{SV}} = \beta + Q_{\text{SV}}^{-1}\delta_{\text{SV}}$. These characterizations do not require specifying whether u causes w or vice versa, or whether z causes x or vice versa, in Figure 1. A causal interpretation as the structural coefficient β requires the corresponding projected bias to vanish. For IV, consistency, exclusion, and exogeneity, together with the homogeneous linear response $y_i(x) = \alpha_i + \beta^\top x$, imply $\mathbb{E}(z_i\varsigma_i) = 0_q$, hence $\delta_{\text{IV}} = 0_p$ and $\beta_{\text{IV}} = \beta$ (Wooldridge, 2010); under heterogeneous effects, the linear IV coefficient may instead have a local causal interpretation under additional conditions (Imbens and Angrist, 1994). For SV, the surrogate controls must capture the confounding-relevant component of x_i, so that $\mathbb{E}[(\Pi_w^\perp x_i)\varsigma_i] = 0_p$, implying $\beta_{\text{SV}} = \beta$. Thus, IV uses instrument-selected variation whereas SV uses residualized variation after surrogate adjustment; when the corresponding causal assumptions fail, the estimators generally target β_{IV} and β_{SV} rather than β.

Further discussion of causal direction, heterogeneous effects, and connections with auxiliary-variable and proximal methods is provided in Appendix B.3.

3. Complementary identification and joint GMM

The IV and SV approaches yield two projected moment restrictions for the same structural coefficient β. We combine these restrictions in a projected GMM system, characterize complementary identification, and derive the estimator's first-order behavior under exact and $n^{-1/2}$-local misspecification.

3.1 Stacked projected moments and identification

Consider the structural model (2.2). Under exact projected validity, the two moment restrictions are

$$\mathbb{E}\{(\Pi_z x_i)\varsigma_i\} = 0_p, \qquad \mathbb{E}\{(\Pi_w^{\perp} x_i)\varsigma_i\} = 0_p.$$

Define the stacked projected regressors and their cross moment with x_i by $h_i := \begin{pmatrix} \Pi_z x_i \\ \Pi_w^{\perp} x_i \end{pmatrix} \in \mathbb{R}^{2p}$, $J := \mathbb{E}(h_i x_i^{\top}) = \begin{pmatrix} Q_{\mathrm{IV}} \\ Q_{\mathrm{SV}} \end{pmatrix}$. The population moment has Jacobian $-J$ with respect to the coefficient vector. The joint restriction is

$$\mathbb{E}\{h_i(y_i - x_i^{\top}\beta)\} = 0_{2p}. \tag{3.1}$$

Proposition 3 (Complementary identification)**.** Suppose that (3.1) holds. Then β is uniquely identified by the stacked projected system if and only if

$$\operatorname{rank}(J) = p \quad \Longleftrightarrow \quad \ker(Q_{\mathrm{IV}}) \cap \ker(Q_{\mathrm{SV}}) = \{0_p\}.$$

Neither block needs to identify the full coefficient separately, provided that their unidentified directions have trivial intersection.

Indeed, equality of the population moments at θ and β is equivalent to $J(\theta-\beta) = 0_{2p}$. Thus, joint identification requires that no nonzero coefficient direction be invisible to both blocks. Neither block needs to identify the full coefficient separately: their unidentified directions need only have trivial intersection. Stacking can therefore restore identification when either block is rank deficient, rather than only improve precision when both are full rank.

Since $\operatorname{rank}(Q_{\text{IV}}) = \operatorname{rank}(\Sigma_{zx}) \leq q$, $q < p$ rules out IV-only identification, but the SV block may recover the missing directions. SV identification depends on $\operatorname{rank}(Q_{\text{SV}})$, without requiring $m \geq p$. The reverse contribution, from IV to directions missing from SV, requires further qualification.

Remark 1 (Scope of complementary identification)**.** If $\mathbb{E}(z_i w_i^\top) = 0_{q\times m}$, the component $\Pi_w x_i$ removed by SV is orthogonal to the instrument span. Hence $\Pi_z x_i = \Pi_z(\Pi_w^\perp x_i)$, and projection contraction gives $0 \preceq Q_{\text{IV}} \preceq Q_{\text{SV}}$. Consequently, $\ker(Q_{\text{SV}}) \subseteq \ker(Q_{\text{IV}})$ and $\operatorname{rank}(J) = \operatorname{rank}(Q_{\text{SV}})$: SV may recover directions missing from IV, but IV cannot recover directions missing from SV.

This restriction applies when randomized assignment is independent of

pre-treatment surrogates. For two rank-deficient blocks to identify the full coefficient jointly, the auxiliary sources must be correlated and their unidentified directions must have trivial intersection. In the CRISPR application, post-perturbation expression surrogates may be correlated with assignment. Such correlation permits, but does not ensure, identification complementarity; causal interpretation still requires both projected moment restrictions to hold.

3.2 Joint estimation under exact specification

Define the sample moment $\widehat{g}(\theta) := n^{-1}\{[X^\top P_Z(Y - X\theta)]^\top, [X^\top P_W^\perp(Y - X\theta)]^\top\}^\top$ and its negative Jacobian $\widehat{J} := n^{-1}\{(X^\top P_Z X)^\top, (X^\top P_W^\perp X)^\top\}^\top$. For a fixed symmetric weight $M \succeq 0$, define

$$\widehat{\beta}(M) := \operatorname*{argmin}_{\theta \in \mathbb{R}^p} \widehat{g}(\theta)^\top M \widehat{g}(\theta). \tag{3.2}$$

Whenever $\widehat{J}^\top M \widehat{J}$ is nonsingular, the minimizer is unique and satisfies

$$\widehat{\beta}(M) = (\widehat{J}^\top M \widehat{J})^{-1} \widehat{J}^\top M \frac{1}{n} \begin{pmatrix} X^\top P_Z Y \\ X^\top P_W^\perp Y \end{pmatrix}. \tag{3.3}$$

For $M \in \mathbb{S}_{++}^{2p}$, population uniqueness is equivalent to $\operatorname{rank}(J) = p$.

If the relevant sample block is nonsingular, the semidefinite weights $\operatorname{diag}(I_p, 0_{p\times p})$ and $\operatorname{diag}(0_{p\times p}, I_p)$ recover $\widehat{\beta}_{\mathrm{IV}}$ and $\widehat{\beta}_{\mathrm{SV}}$, respectively. Section 4

also characterizes positive-definite weights inducing their population sensitivities. When $q > p$, projection may discard overidentifying information from the raw IV moment.

Because the sample moments use estimated projection operators, their first-order terms generally differ from $h_i\varsigma_i$. Define the terms at the structural coefficient β by

$$\begin{aligned}\psi_{\mathrm{IV},i} &= (\Pi_z x_i)\varsigma_i + (x_i - \Pi_z x_i) z_i^\top \Sigma_z^{-1} \mathbb{E}(z_i\varsigma_i), \\ \psi_{\mathrm{SV},i} &= (\Pi_w^\perp x_i)\{\varsigma_i - w_i^\top \Sigma_w^{-1} \mathbb{E}(w_i\varsigma_i)\}. \end{aligned} \tag{3.4}$$

When an individual projected system is identified and valid, its statistical target equals β, and the corresponding term agrees with the single-source score defined earlier. These definitions remain applicable when an individual block is rank deficient.

Under the raw IV restriction $\mathbb{E}(z_i\varsigma_i) = 0_q$, the IV projection correction vanishes. The SV term involves the disturbance residualized on w_i, reflecting estimation of the surrogate projection. Write $\psi_i := (\psi_{\mathrm{IV},i}^\top, \psi_{\mathrm{SV},i}^\top)^\top$, $\Sigma := \mathrm{Var}(\psi_i)$,

$$K_M := (J^\top M J)^{-1} J^\top M \tag{3.5}$$

, and $V(M) := K_M \Sigma K_M^\top$.

Assumption 3 (Combined GMM conditions)**.** Suppose that the observations are i.i.d., the dimensions p, q, m are fixed, and the maintained second-moment conditions of Section 2 hold.

(i) **Projected validity and identification.** The restrictions in (3.1) hold, $\Sigma_z \in \mathbb{S}^q_{++}$, and $\Sigma_w \in \mathbb{S}^m_{++}$. The fixed symmetric weight $M \succeq 0$ satisfies $J^\top M J \in \mathbb{S}^p_{++}$.

(ii) **Score moments and nondegeneracy.** Suppose $\mathbb{E}\|x_i z_i^\top\|_{\mathrm{op}}^2 < \infty$, $\mathbb{E}\|z_i\{\varsigma_i - z_i^\top \Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i)\}\|_2^2 < \infty$, $\mathbb{E}\|x_i\varsigma_i\|_2^2 < \infty$, $\mathbb{E}\|x_i w_i^\top\|_{\mathrm{op}}^2 < \infty$, and $\mathbb{E}\|w_i\{\varsigma_i - w_i^\top \Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i)\}\|_2^2 < \infty$. Further, $V(M) \in \mathbb{S}^p_{++}$.

The covariance Σ may be singular because the system can contain redundant score directions. Positive definiteness of $V(M)$ ensures nondegenerate coefficient inference. Positive definiteness of Σ is required when its inverse is used for efficient weighting.

Theorem 4 (Identification, consistency, and asymptotic normality under exact specification)**.** Under Assumption 3 (i), the following statements hold. (i) β is the unique solution to the population stacked moment restriction. (ii) $\widehat{\beta}(M) \xrightarrow{\mathrm{p}} \beta$. (iii) If Assumption 3 (ii) also holds, then $\sqrt{n}\{\widehat{\beta}(M) - \beta\} \xrightarrow{\mathrm{d}} \mathcal{N}\{0, V(M)\}$.

3.3 Inference under local misspecification

We next consider a triangular array with observations that are i.i.d. within each row. The distribution of (x_i, z_i, w_i), the coefficient β, and the dimensions p, q, m are fixed; only the conditional distribution of ς_i may depend on n. Thus, the population projections, J, and K_M are fixed. The design distribution is held fixed, rather than the realized sample design.

Write $\mathbb{E}_n$ and Var_n for expectation and variance under the row-n distribution. Following Armstrong and Kolesár (2021), impose

$$\mathbb{E}_n(h_i\varsigma_i) = \frac{\tau}{\sqrt{n}}, \qquad \tau = (\tau_z^\top, \tau_w^\top)^\top \in \mathbb{R}^{2p}. \tag{3.6}$$

The vector τ is fixed and must be attainable under the maintained design distribution. For example, if $a^\top h_i = 0$ almost surely, then necessarily $a^\top \tau = 0$.

In this regime, the score definitions in (3.4) retain the same form, with the disturbance cross moments replaced by $\mathbb{E}_n(z_i\varsigma_i)$ and $\mathbb{E}_n(w_i\varsigma_i)$. Since the distribution of (x_i, z_i, w_i) is fixed, the population projection operators Π_z and $\Pi_w^\perp$ remain unchanged. Projection orthogonality ensures that the correction terms have zero expectation, giving $\mathbb{E}_n(\psi_{\mathrm{IV},i}) = \mathbb{E}_n\{(\Pi_z x_i)\varsigma_i\} = \tau_z/\sqrt{n}$ and $\mathbb{E}_n(\psi_{\mathrm{SV},i}) = \mathbb{E}_n\{(\Pi_w^\perp x_i)\varsigma_i\} = \tau_w/\sqrt{n}$. Hence $\mathbb{E}_n(\psi_i) = \tau/\sqrt{n}$. Estimating the projections can therefore affect the first-order covariance,

but does not alter the local deviations of the projected moments from zero.

Assumption 4 (Combined GMM conditions under local misspecification)**.** Under the fixed population design distribution described above, suppose the following conditions hold.

(i) **Local restriction and identification.** Equation (3.6) holds for a fixed, attainable τ. The design has finite second moments, $\Sigma_z \in \mathbb{S}^q_{++}$, and $\Sigma_w \in \mathbb{S}^m_{++}$. The fixed symmetric weight $M \succeq 0$ satisfies $J^\top M J \in \mathbb{S}^p_{++}$.

(ii) **Moment conditions.** Suppose $\mathbb{E}\|x_i x_i^\top\|_{\text{op}}^2 < \infty$, $\mathbb{E}\|z_i z_i^\top\|_{\text{op}}^2 < \infty$, $\mathbb{E}\|w_i w_i^\top\|_{\text{op}}^2 < \infty$, $\mathbb{E}\|x_i z_i^\top\|_{\text{op}}^2 < \infty$, and $\mathbb{E}\|x_i w_i^\top\|_{\text{op}}^2 < \infty$. Further, suppose $\sup_n \mathbb{E}_n(\varsigma_i^2) < \infty$, $\sup_n \mathbb{E}_n\|x_i\varsigma_i\|_2^2 < \infty$, $\sup_n \mathbb{E}_n\|z_i\{\varsigma_i - z_i^\top \Sigma_z^{-1}\mathbb{E}_n(z_i\varsigma_i)\}\|_2^2 < \infty$, and $\sup_n \mathbb{E}_n\|w_i\{\varsigma_i - w_i^\top \Sigma_w^{-1}\mathbb{E}_n(w_i\varsigma_i)\}\|_2^2 < \infty$.

(iii) **Score tails, covariance convergence, and nondegeneracy.** For some fixed $\eta > 0$, the scores satisfy $\sup_n \mathbb{E}_n\|\psi_i\|_2^{2+\eta} < \infty$. The score covariance satisfies $\text{Var}_n(\psi_i) \to \Sigma_{\text{miss}} \in \mathbb{S}^{2p}_+$, and $V_{\text{miss}}(M) := K_M \Sigma_{\text{miss}} K_M^\top \in \mathbb{S}^p_{++}$.

The moment conditions support sample moment convergence and the projection expansions, while the uniform $2+\eta$ score moment bound ensures the Lindeberg condition.

Theorem 5 (Combined GMM under local misspecification)**.** Under Assumption 4 (i), the following statements hold. (i) The population GMM criterion has the unique minimizer $\beta_{M,n} = \beta + n^{-1/2}K_M\tau$, and $\beta_{M,n} \to \beta$. (ii) If Assumption 4 (ii) also holds, then $\widehat{\beta}(M) \xrightarrow{\mathrm{p}} \beta$. (iii) Under the full assumption, $\sqrt{n}\{\widehat{\beta}(M) - \beta\} \xrightarrow{\mathrm{d}} \mathcal{N}\{K_M\tau, V_{\mathrm{miss}}(M)\}$.

The corresponding single-source conclusions hold under the same conditions when the selected block identifies the full coefficient.

Under exact specification, weighting affects first-order variance. Under local misspecification, it also determines the local bias $K_M\tau$. The two regimes share the same form of asymptotic linear representation, although their covariance limits need not coincide. Setting $\tau = 0$ and $\Sigma_{\mathrm{miss}} = \Sigma$ recovers the exact-specification limit.

4. Optimal weighting and sensitivity analysis

We choose the weight using squared Euclidean loss. Since first-order behavior depends on M through K_M, the optimization can be expressed directly in terms of the sensitivity.

4.1 Asymptotic risk under local misspecification

For a fixed local sequence indexed by τ, define

$$\mathrm{AMSE}_\tau(M) := \lim_{n\to\infty} n\,\mathbb{E}_\tau\|\widehat{\beta}(M) - \beta\|_2^2, \tag{4.1}$$

whenever the limit exists. Here $\mathbb{E}_\tau$ denotes expectation along the specified local sequence. If the squared root-n estimation errors are uniformly integrable, Theorem 5 gives

$$\mathrm{AMSE}_\tau(M) = \|K_M\tau\|_2^2 + \mathrm{tr}(K_M\Sigma_{\mathrm{miss}}K_M^\top).$$

Let $D := \mathrm{diag}(c_z I_p, c_w I_p)$, with $c_z, c_w \geq 0$, and $\mathcal{C}(D) := \{Du : \|u\|_2 \leq 1\}$. The radii are prespecified sensitivity parameters on the root-n projected-moment scale. When both radii are positive, the set imposes the joint budget $\|\tau_z\|_2^2/c_z^2 + \|\tau_w\|_2^2/c_w^2 \leq 1$. A zero radius sets the corresponding block's violation to zero. The radii must be specified relative to the units of the moments.

For comparison across local directions, consider sequences sharing the same fixed design distribution and the same limiting score covariance Σ_{miss}. The latter is an additional restriction on the sequence family; it does not follow solely from fixing the design distribution.

Lemma 6 (Worst-case asymptotic risk)**.** For $K \in \mathbb{R}^{p\times 2p}$ satisfying $KJ =$

I_p, recall its first-order risk by $\mathrm{AMSE}_\tau(K) := \|K\tau\|_2^2 + \mathrm{tr}(K\Sigma_{\mathrm{miss}}K^\top)$. Then

$$R(K) := \sup_{\tau \in \mathcal{C}(D)} \mathrm{AMSE}_\tau(K) = \|KD\|_{\mathrm{op}}^2 + \mathrm{tr}(K\Sigma_{\mathrm{miss}}K^\top). \tag{4.2}$$

For an estimator with the stated local asymptotic distribution and sensitivity K, this first-order risk equals the limit in (4.1) when its squared root-n errors are uniformly integrable.

This criterion takes the supremum of pointwise asymptotic risks; it does not interchange a supremum and a finite-sample risk limit. If $\mathcal{C}(D)$ contains unattainable directions, $R(K)$ is a conservative envelope for the worst-case risk over the attainable directions.

4.2 Risk-optimal weighting

The risk-optimal sensitivity solves

$$K^* \in \operatorname*{argmin}_{K:\, KJ = I_p} \left\{ \|KD\|_{\mathrm{op}}^2 + \mathrm{tr}(K\Sigma_{\mathrm{miss}}K^\top) \right\}. \tag{4.3}$$

The constraint is affine and the objective is convex. If $\Sigma_{\mathrm{miss}} \in \mathbb{S}_{++}^{2p}$, the objective is strongly convex and coercive, giving a unique minimizer whenever $\mathrm{rank}(J) = p$.

The positive-definite covariance condition used for uniqueness and feasible optimal weighting is stronger than the conditions required for complementary identification and fixed-weight inference. In particular, if Q_{sv}

is singular, some nonzero $a \in \ker(Q_{\mathrm{sv}})$ satisfies $a^\top \Pi_w^\perp x_i = 0$ almost surely, and hence $a^\top \psi_{\mathrm{sv},i} = 0$ along the local sequence. Thus Σ_{miss} is singular, even though $\mathrm{rank}(J) = p$ may still hold. Such cases remain covered by the preceding identification and fixed-weight asymptotic results.

The optimization is equivalent to optimizing over positive-definite GMM weights. Every GMM sensitivity satisfies $K_M J = I_p$. Let $N_J \in \mathbb{R}^{2p \times p}$ have orthonormal columns spanning $\ker(J^\top)$. For any $KJ = I_p$, the positive-definite weights inducing K are exactly

$$\mathcal{M}(K) := \{K^\top A K + N_J B N_J^\top : A, B \in \mathbb{S}_{++}^p\}.$$

Each member is positive definite because $\ker(K) \cap \mathcal{C}(J) = \{0_{2p}\}$. Moreover, $J^\top M J = A$ and $J^\top M = AK$, so $(J^\top M J)^{-1} J^\top M = K$.

Conversely, for any positive-definite weight M inducing K, set $A := J^\top M J$. Then $A \in \mathbb{S}_{++}^p$, $J^\top M = AK$, and $M - K^\top A K = M - MJ(J^\top M J)^{-1} J^\top M$. Factoring out $M^{1/2}$ shows that this remainder is positive semidefinite with null space $\mathcal{C}(J)$. It therefore equals $N_J B N_J^\top$ for some $B \in \mathbb{S}_{++}^p$, establishing the representation. The weights inducing the unique optimum are consequently $\mathcal{M}(K^*)$.

Under exact specification, take $D = 0$ and use the exact covariance Σ. When $\Sigma \in \mathbb{S}_{++}^{2p}$, the unique variance-optimal sensitivity is $K^* = (J^\top \Sigma^{-1} J)^{-1} J^\top \Sigma^{-1}$. The efficient-GMM weight $M^\star = \Sigma^{-1}$ is a canoni-

cal inducing weight (Hansen, 1982). Under local misspecification, the term $\|KD\|_{\text{op}}^2$ trades sensitivity to moment violations against precision (Armstrong and Kolesár, 2021).

When $Q_{\text{IV}} \in \mathbb{S}_{++}^p$, the IV-only sensitivity is $K^{(\text{IV})} = (Q_{\text{IV}}^{-1}, 0_{p\times p})$, with risk $R\{K^{(\text{IV})}\} = c_z^2\|Q_{\text{IV}}^{-1}\|_{\text{op}}^2 + \text{tr}(Q_{\text{IV}}^{-1}\Sigma_{\text{IV,miss}}Q_{\text{IV}}^{-1})$. Similarly, when $Q_{\text{SV}} \in \mathbb{S}_{++}^p$, the SV-only sensitivity is $K^{(\text{SV})} = (0_{p\times p}, Q_{\text{SV}}^{-1})$, with risk $R\{K^{(\text{SV})}\} = c_w^2\|Q_{\text{SV}}^{-1}\|_{\text{op}}^2 + \text{tr}(Q_{\text{SV}}^{-1}\Sigma_{\text{SV,miss}}Q_{\text{SV}}^{-1})$. The covariance blocks are the corresponding diagonal blocks of Σ_{miss}.

4.3 Feasible estimation

The risk-optimal rule can be implemented by replacing the population Jacobian and score covariance with consistent sample analogues. Choose a fixed first-step weight $M_0 \in \mathbb{S}_{++}^{2p}$, for example $M_0 = I_{2p}$, and compute the preliminary estimator $\widehat{\beta}(M_0)$. Let $\widehat{r} := Y - X\widehat{\beta}(M_0)$ denote the preliminary residual vector, with ith component $\widehat{r}_i = y_i - x_i^\top\widehat{\beta}(M_0)$. The plug-in first-order scores are

$$\widehat{\psi}_{\text{IV},i} := X^\top Z(Z^\top Z)^{-1}z_i\widehat{r}_i + \left\{x_i - X^\top Z(Z^\top Z)^{-1}z_i\right\} z_i^\top(Z^\top Z)^{-1}Z^\top\widehat{r},$$

$$\widehat{\psi}_{\text{SV},i} := \left\{x_i - X^\top W(W^\top W)^{-1}w_i\right\} \times \left\{\widehat{r}_i - w_i^\top(W^\top W)^{-1}W^\top\widehat{r}\right\}.$$

Write $\widehat{\psi}_i := (\widehat{\psi}_{\text{IV},i}^\top, \widehat{\psi}_{\text{SV},i}^\top)^\top$, and estimate the limiting score covariance by

$$\widehat{\Sigma} := \frac{1}{n}\sum_{i=1}^n(\widehat{\psi}_i - \overline{\widehat{\psi}})(\widehat{\psi}_i - \overline{\widehat{\psi}})^\top, \qquad \overline{\widehat{\psi}} := \frac{1}{n}\sum_{i=1}^n\widehat{\psi}_i.$$

Retaining the prespecified D, define the feasible risk-optimal sensitivity by

$$\widehat{K}^* \in \underset{K:\, K\widehat{J}=I_p}{\operatorname{argmin}} \left\{ \|KD\|_{\mathrm{op}}^2 + \operatorname{tr}(K\widehat{\Sigma}K^\top) \right\}.$$

The corresponding estimator is

$$\widehat{\beta}_{\mathrm{opt}} = \widehat{K}^* \frac{1}{n} \begin{pmatrix} X^\top P_Z Y \\ X^\top P_W^\perp Y \end{pmatrix}.$$

Thus the oracle bias–variance tradeoff is implemented by estimating the Jacobian and first-order score covariance and solving the sample analogue of (4.3).

Equivalently, on the event that $\operatorname{rank}(\widehat{J}) = p$, let $\widehat{N}_J \in \mathbb{R}^{2p\times p}$ have orthonormal columns spanning $\ker(\widehat{J}^\top)$. A canonical positive-definite GMM weight inducing $\widehat{K}^*$ is $\widehat{M}_{\mathrm{opt}} := \widehat{K}^{*\top}\widehat{K}^* + \widehat{N}_J\widehat{N}_J^\top$. Indeed, $(\widehat{J}^\top\widehat{M}_{\mathrm{opt}}\widehat{J})^{-1}\widehat{J}^\top\widehat{M}_{\mathrm{opt}} = \widehat{K}^*$, so that $\widehat{\beta}(\widehat{M}_{\mathrm{opt}}) = \widehat{\beta}_{\mathrm{opt}}$. More generally, any sample analogue of $\mathcal{M}(\widehat{K}^*)$ yields the same estimator.

Proposition 7 (Feasible risk-optimal weighting)**.** Suppose Assumption 4 holds, $\Sigma_{\mathrm{miss}} \in \mathbb{S}_{++}^{2p}$, and D is fixed. Let K^* denote the unique solution of (4.3). Then $\widehat{J} \xrightarrow{\mathrm{p}} J$, $\widehat{\Sigma} \xrightarrow{\mathrm{p}} \Sigma_{\mathrm{miss}}$, and $\widehat{K}^* \xrightarrow{\mathrm{p}} K^*$. Moreover, $\sqrt{n}\{\widehat{\beta}_{\mathrm{opt}} - \beta\} \xrightarrow{\mathrm{d}} \mathcal{N}\left(K^*\tau,\, K^*\Sigma_{\mathrm{miss}}K^{*\top}\right)$. Hence estimating the risk-optimal sensitivity does not alter its first-order oracle limit.

Remark 2 (Exact specification)**.** Under exact specification, with $D = 0$ and $\Sigma_{\text{miss}} = \Sigma$, the bias term vanishes. Risk minimization therefore reduces to minimizing $\text{tr}(K\Sigma K^\top)$, the trace of the asymptotic covariance, subject to $KJ = I_p$. The unique minimizer is $K^* = (J^\top\Sigma^{-1}J)^{-1}J^\top\Sigma^{-1}$, and $\sqrt{n}\{\widehat{\beta}_{\text{opt}} - \beta\} \xrightarrow{\text{d}} \mathcal{N}\left(0, (J^\top\Sigma^{-1}J)^{-1}\right)$. Thus, the feasible risk-optimal procedure reduces to efficient two-step projected GMM when the sensitivity budget is zero.

4.4 Risk comparisons and confidence intervals

The single-source sensitivities are feasible for the combined optimization whenever their respective blocks identify the full coefficient.

Proposition 8 (Risk dominance)**.** Suppose $\text{rank}(J) = p$, $Q_{\text{IV}}, Q_{\text{SV}} \in \mathbb{S}^p_{++}$, and $\Sigma_{\text{miss}} \in \mathbb{S}^{2p}_{++}$. For the same set $\mathcal{C}(D)$ and covariance Σ_{miss}, let K^* solve (4.3). Then $R(K^*) \leq \min\{R(K^{(\text{IV})}), R(K^{(\text{SV})})\}$.

Under exact specification, with $D = 0$ and $\Sigma_{\text{miss}} = \Sigma$, $R(K) = \text{tr}(K\Sigma K^\top)$ and $K^* = (J^\top\Sigma^{-1}J)^{-1}J^\top\Sigma^{-1}$. Consequently,

$$\text{tr}\{(J^\top\Sigma^{-1}J)^{-1}\} \leq \min\left\{\text{tr}(Q_{\text{IV}}^{-1}\Sigma_{\text{IV}}Q_{\text{IV}}^{-T}), \text{tr}(Q_{\text{SV}}^{-1}\Sigma_{\text{SV}}Q_{\text{SV}}^{-T})\right\}.$$

Thus variance dominance is the zero-bias special case of risk dominance.

When all directions in $\mathcal{C}(D)$ are attainable, the corresponding asymptotic distributions hold, and squared root-n errors are uniformly integrable,

these inequalities compare worst-case local AMSE. Under exact specification, Assumption 3 with $M^\star = \Sigma^{-1}$, together with uniform integrability, yields the AMSE comparison. Without these conditions, the inequalities remain comparisons of first-order risk envelopes.

The inequalities follow from $K^{(\mathrm{IV})}J = K^{(\mathrm{SV})}J = I_p$. They compare oracle worst-case risks over the same $\mathcal{C}(D)$, without guaranteeing smaller risk at each fixed τ or smaller finite-sample MSE. For the feasible estimator, MSE convergence additionally requires uniform integrability of squared root-n errors.

Comparison with IV alone requires only Q_{IV} to be nonsingular; comparison with SV alone requires only Q_{SV}. Combined identification requires $\mathrm{rank}(J) = p$, even when individual blocks are singular.

Remark 3 (Confidence intervals allowing local misspecification)**.** For a prespecified nonzero $a \in \mathbb{R}^p$, the bias envelope is $\sup_{\tau \in \mathcal{C}(D)} |a^\top K\tau| = \|DK^\top a\|_2$. For the feasible estimator, let $\widehat{b}_a := \|D\widehat{K}^\top a\|_2$ and $\widehat{s}_a^2 := a^\top \widehat{K}\widehat{\Sigma}\widehat{K}^\top a$.

Suppose $\widehat{K} \xrightarrow{\mathrm{p}} K$, $\widehat{\Sigma} \xrightarrow{\mathrm{p}} \Sigma_{\mathrm{miss}}$, $a^\top K\Sigma_{\mathrm{miss}}K^\top a > 0$, and the estimator has the stated local asymptotic normal distribution. Then the interval

$$a^\top \widehat{\beta} \;\pm\; \mathrm{cv}_\alpha\left(\frac{\widehat{b}_a}{\widehat{s}_a}\right)\frac{\widehat{s}_a}{\sqrt{n}}$$

has asymptotic coverage at least $1-\alpha$ along each fixed attainable local se-

quence with $\tau \in \mathcal{C}(D)$. Here, $\mathrm{cv}_\alpha(t)$ denotes the critical value of the folded normal distribution: for $Z \sim \mathcal{N}(t, 1)$, $\mathrm{cv}_\alpha(t)$ is the $(1-\alpha)$-quantile of $|Z|$, that is, $\Pr\{|Z| \le \mathrm{cv}_\alpha(t)\} = 1 - \alpha$. Equivalently, $\mathrm{cv}_\alpha(t)$ is the solution c to $\Phi(c - t) - \Phi(-c - t) = 1 - \alpha$, where Φ denotes the standard normal distribution function. In particular, $\mathrm{cv}_\alpha(0) = z_{1-\alpha/2}$ (Armstrong and Kolesár, 2021). Uniform coverage additionally requires uniform asymptotic approximation and uniform consistency of the estimated ingredients. When $D = 0$, the interval reduces to the Wald interval.

Appendix D examines a one-parameter weighted least-squares (WLS) path connecting the IV-only and SV-only estimators as a restricted GMM weighting family. It characterizes the finite-sample conditional risk along this path and provides an oracle surrogate criterion with a risk guarantee under bounded condition numbers.

5. Simulation studies

We consider two scalar designs illustrating the finite-sample implications of the weighting results. The first studies efficiency under correct specification, while the second introduces local misspecification in both the IV and SV moments and examines the resulting bias–variance tradeoff. OLS is included as a benchmark. All projections and data-dependent GMM

weights are re-estimated in each of 1,000 Monte Carlo replications, with $n \in \{200, 600, 1000, 1400, 1800\}$.

5.1 Efficiency under exact specification

Consider $z_i = s_i + \eta_i$, $w_i = (u_i, s_i)^\top$, and $x_i = u_i + 2s_i + v_i$, with

$$y_i = \beta x_i + u_i + \varepsilon_i,$$

where $s_i, \eta_i, u_i, v_i, \varepsilon_i$ are mutually independent standard normal variables. The IV-only and SV-only estimators both have asymptotic variance 1, compared with 0.68 for equal-weight GMM and 0.50 for efficient GMM. We report MSE at $\beta = 1$, and two-sided 5% Wald rejection probabilities under $H_0 : \beta = 0$ and the local alternative $\beta = 2/\sqrt{n}$.

Figure 2 confirms the gains from combining the two moment blocks: efficient GMM attains the lowest MSE and highest local power among size-controlled procedures, while equal-weight GMM also improves on either single-source estimator. OLS remains biased because x_i contains u_i; its bias is 1/6, producing persistent MSE and severe size distortion.

5.2 Risk under local misspecification

We retain the design in Section 5.1 but replace the outcome equation with

$$y_i = \beta x_i + u_i + \varepsilon_i + \tau_z z_i/(2\sqrt{n}) + \tau_w v_i/\sqrt{n}$$

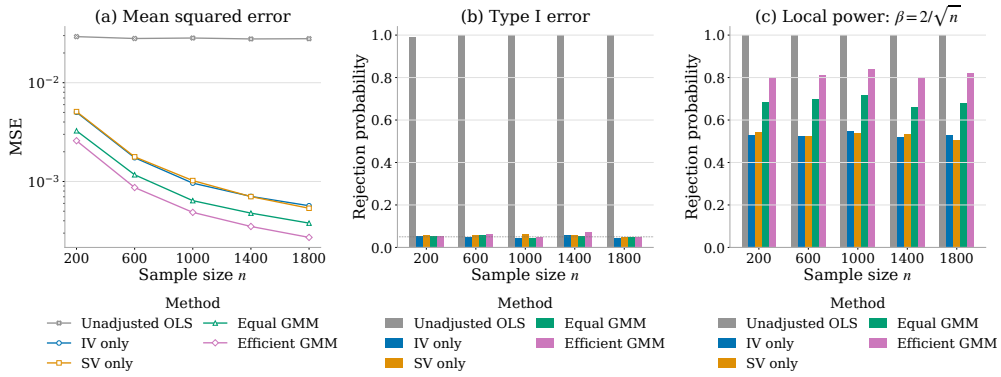


Figure 2: Performance under correct specification. Panel (a) reports MSE at $\beta = 1$ on a logarithmic scale; panels (b) and (c) report rejection probabilities under $H_0 : \beta = 0$ and $\beta = 2/\sqrt{n}$, respectively.

. Since the population projected regressors are $h_i = (z_i, v_i)^\top$, this gives $\sqrt{n}\,\mathbb{E}\{h_i(y_i - \beta x_i)\} = (\tau_z, \tau_w)^\top$. Following Lemma 6, set $D = \mathrm{diag}(1, 1)$ and $\mathcal{C}(D) = \{(\tau_z, \tau_w)^\top : \tau_z^2 + \tau_w^2 \leq 1\}$, allowing both moments to deviate.

We compare OLS, IV only, SV only, equal-weight GMM, and feasible local-risk GMM over 16 equally spaced boundary directions. At $\beta = 1$, we report worst-case MSE and Type I error for $H_0 : \beta = 1$, together with minimum local power under $\beta = 1 + 2/\sqrt{n}$. OLS uses conventional Wald inference; the remaining methods use the misspecification-robust intervals in Remark 3, all at the 5% level.

Across the reported sample sizes, feasible local-risk GMM has the lowest

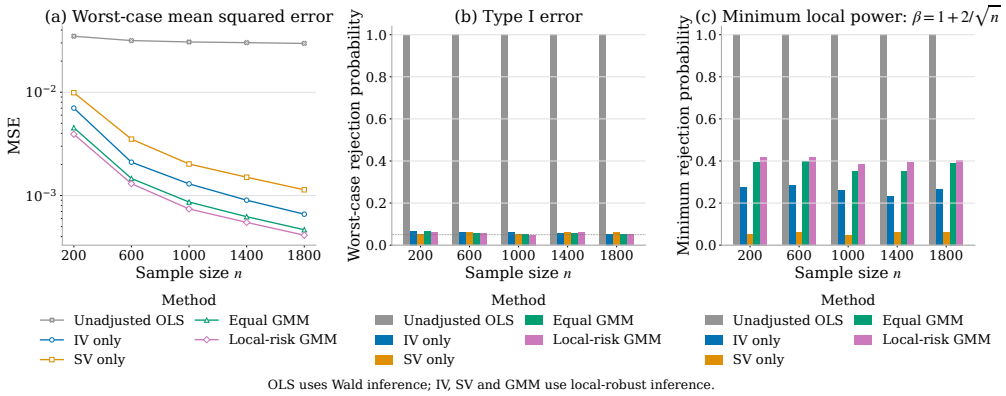


Figure 3: Performance under local misspecification of both the IV and SV moments. Panel (a) reports worst-case MSE on a logarithmic scale; panels (b) and (c) report worst-case Type I error and minimum local power over the evaluated boundary directions.

estimated worst-case MSE among the methods compared (Figure 3). This pattern agrees with the oracle first-order risk comparison in Proposition 8. The robust procedures maintain approximately nominal size, with some finite-sample overrejection, whereas OLS remains distorted. Local misspecification substantially reduces single-source power, especially for SV only, while the combined procedures retain more power. Local-risk GMM has higher minimum local power than equal-weight GMM here, although the former optimizes worst-case estimation risk rather than testing power.

6. CRISPR perturbation analysis

6.1 Data, estimand, and estimation

We apply the proposed methods to the K562 essential-scale CRISPRi Perturb-seq experiment of Replogle et al. (2022). In this experiment, paired single-guide RNAs repress individual target genes, and single-cell RNA sequencing with direct guide capture jointly measures gene expression and perturbation identity six days after infection. We analyze pseudobulk expression profiles constructed within perturbation–batch groups across 48 sequencing batches. After filtering, the analysis includes 213 perturbation targets and a common panel of 2,000 candidate downstream genes. Each target-specific analysis combines 30–48 perturbed profiles with 48 control profiles, giving $n = 78$–96. Details of pseudobulk construction, gene selection, and preprocessing are provided in Appendix E.

For each perturbation–outcome pair, we fit (2.1) with x_i and y_i denote target and downstream gene expression, respectively, and z_i indicates perturbation versus control. The surrogate controls w_i are the first 10 principal component scores of a common panel of 1,000 genes selected for low differential-expression responses across perturbations. The perturbation indicator provides the IV channel, whereas residualization on w_i provides the

SV channel. We compare the IV, SV, and optimally weighted two-step GMM estimators over $213 \times 2,000 = 426,000$ pairs, with standard errors clustered by batch. We use the feasible two-step GMM estimator described in Remark 2, corresponding to the $D = 0$ risk-optimal rule. Keeping its fitted weights and point estimates fixed, we compare conventional Wald intervals with local misspecification-adjusted intervals constructed as in Remark 3 for a prespecified moment-deviation set.

6.2 Results and sensitivity analysis

For the local sensitivity analysis, we specify the moment-deviation set as

$$\sqrt{n}\,\mathbb{E}\{g_i(\beta)\} = \tau, \qquad \tau = \mathrm{diag}(r_{\mathrm{IV}}, r_{\mathrm{SV}})u, \qquad \|u\|_2 \le 1,$$

where $g_i(\beta)$ is the implemented moment vector defined in Appendix E, $r_j = 0.5\sqrt{\widehat{\Omega}_{jj}}$, and $\widehat{\Omega}$ is the batch-clustered covariance of the stacked moment scores evaluated at the GMM estimate. The same pair-specific deviation set is used for all three estimators. For each perturbation, method, and inference rule, we apply the Benjamini–Hochberg procedure across the 2,000 outcomes at a nominal false discovery rate of 5%.

Figure 4 shows substantial overlap among the three methods, together with method-specific discoveries. Wald GMM identifies 84,363 significant pairs, including some not selected by either single-source estimator. Lo-

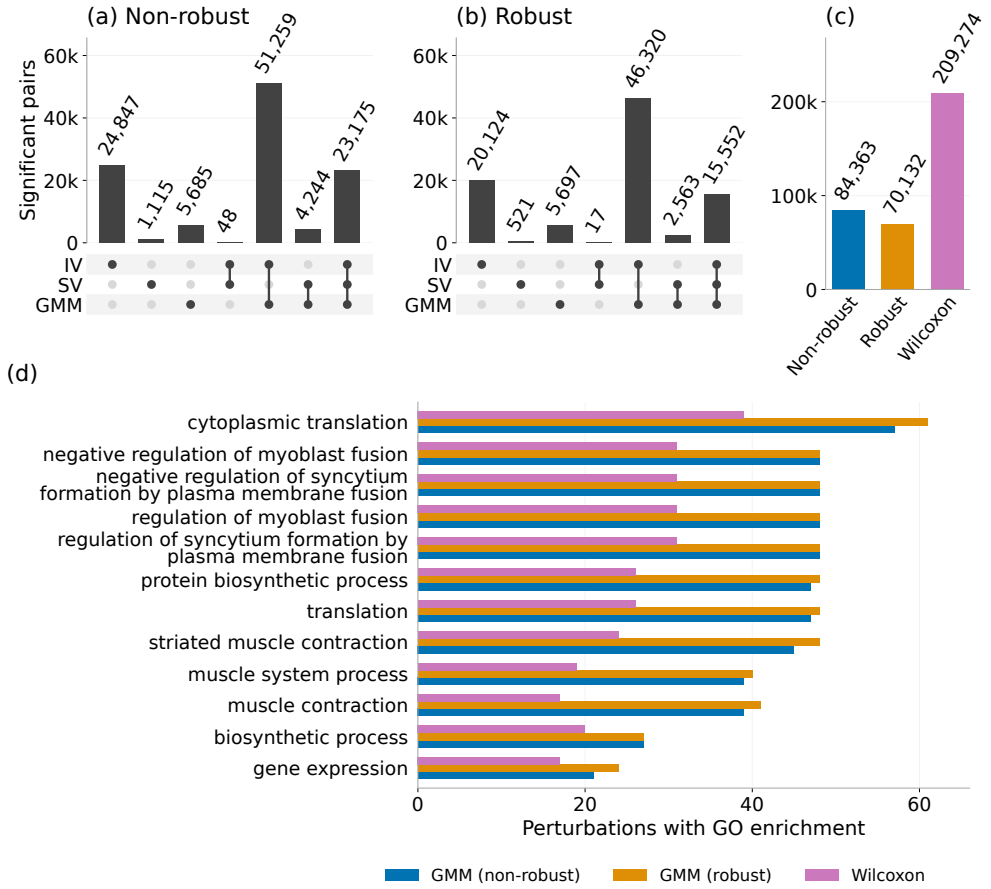


Figure 4: Discovery overlap and GO biological-process enrichment across 213 perturbations. (a,b) Exclusive intersections under non-robust and robust inference. Counts represent significant perturbation-outcome gene pairs; outcomes found for different perturbations are counted separately. Filled dots mark included methods and pale dots mark excluded methods. The IV/SV intersection excluding GMM contains 48 and 17 pairs, respectively. Zero-effect tests use BH q-values at most 0.05 within each perturbation and method over 2,000 outcomes. (c) GMM and batch-stratified Wilcoxon discovery counts. (d) The 12 GO terms with the greatest summed numbers of enriched perturbations across selections. Enrichment uses the tested outcomes as background and BH q-values at most 0.05 within each perturbation and selection.

cal misspecification-adjusted inference retains 70,132 GMM discoveries, a reduction of approximately 16.9%. Thus, most GMM discoveries persist under the chosen calibration of local moment deviations.

The batch-stratified Wilcoxon analysis identifies 209,274 significant pairs on the same testing panel. This comparison concerns different statistical targets: Wilcoxon detects perturbation-associated expression differences, whereas GMM tests the target-expression coefficient under the maintained moment restrictions. Despite its smaller discovery set, GMM yields recurrent GO biological-process enrichment, particularly for translation- and biosynthesis-related processes. These main biological themes remain similar after adjustment for local misspecification.

7. Discussion

This paper provides a common projected-moment framework for deconfounding with instruments and surrogate controls. Under valid moment conditions, their distinct sources of exposure variation can jointly identify the structural coefficient even when neither source does so alone. Combining these moments through GMM can improve precision under exact specification and balance bias against variance under local misspecification.

A causal interpretation of the combined estimator requires assump-

tions supporting the validity of both moment restrictions. Even when the combined system identifies every coefficient, weak identifying variation can lead to imprecise estimates. The efficiency comparisons concern the projected moments, which may discard information from the original IV moments. Complementary identification and fixed-weight inference allow singular score covariances, whereas the current feasible risk-optimal weighting theory requires positive definiteness. It therefore excludes cases where both sources are rank deficient but jointly identify the coefficient, since a singular Q_{sv} implies a singular stacked score covariance. Extending feasible optimal weighting to these cases remains an open problem. The local sensitivity analysis quantifies departures within a prespecified set; it does not establish the validity of either source from the data.

In the CRISPR application, combined GMM identifies significant target–outcome gene pairs not selected by either single-source estimator. Most GMM discoveries persist under the chosen local sensitivity calibration. The GMM analysis concerns target-expression coefficients, while differential-expression analysis examines expression differences between perturbed and control cells. The selected gene sets show recurrent enrichment for translation- and biosynthesis-related processes.

Nonlinear structural models provide one extension, since IV and SV mo-

ments may capture different features of the exposure–outcome relationship. Combining them would require defining a common target and establishing valid moment restrictions. Basis expansions offer a possible approach, although approximation error and the interpretation of the resulting target require further study. Measurement error in exposures and surrogate controls poses another challenge because it can alter the projections and the validity of the moment restrictions. Corrected-moment GMM methods (Evdokimov and Zeleneev, 2025) suggest a possible approach, subject to assumptions on the measurement error mechanism. How these corrections affect complementary identification, weighting, and inference remains an open problem.

Supplementary Material

The supplementary material contains additional discussion of related work, proofs of all theoretical results, and further details on simulation and real-data analysis.

Acknowledgements

The corresponding author acknowledges support from the National Natural Science Foundation of China (No. 12601487), and the Institute of Data

Science (IDS) at the University of Hong Kong through the HKU100 award.

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Supplemental material

The supplementary material is organized as follows. Section A summarizes the notation. Section B provides additional discussion of the projected IV formulation, surrogate-control residualization, and the assumptions required for causal interpretation. Section C contains proofs of the main theoretical results. Section D studies a restricted weighted least-squares path connecting the IV and SV estimators, establishing its GMM representation, finite-sample conditional risk bounds, and an oracle surrogate criterion. Section E describes the data preparation and implementation of the real-data analysis, including surrogate construction, estimation, batch-clustered inference, local sensitivity analysis, multiple testing, and biological enrichment. We use lower-case letters for unit-level variables and upper-case letters for sample arrays: $y_i \in \mathbb{R}$, $x_i \in \mathbb{R}^p$, $z_i \in \mathbb{R}^q$, $w_i \in \mathbb{R}^m$, and $u_i \in \mathbb{R}^r$ denote the outcome, regressors, instruments, surrogate controls, and latent confounders, respectively, with Y, X, Z, W, U denoting their corresponding sample arrays.

A. Notation

Population L_2 projection operators onto the linear spans of z and w are denoted by Π_z and Π_w, respectively, with $\Pi_w^\perp := I - \Pi_w$ denoting the corresponding residual operator, where I is the identity operator. For any sample matrix A with n rows, $\mathcal{C}(A)$ denotes its column space, P_A the Euclidean projection onto $\mathcal{C}(A)$, and $P_A^\perp := I_n - P_A$ the corresponding residual-maker, where I_n is the $n \times n$ identity matrix. For any matrix A, $\ker(A) := \{v : Av = 0\}$ denotes its

null space and $\operatorname{rank}(A) := \dim \mathcal{C}(A)$ its rank.

We use $\|\cdot\|_2$, $\|\cdot\|_{\mathrm{F}}$, and $\|\cdot\|_{\mathrm{op}}$ for the Euclidean, Frobenius, and operator norms, respectively. We write $\operatorname{tr}(A)$ for the trace of a square matrix A, and diag for a diagonal or block-diagonal matrix formed from its arguments. For a symmetric matrix A, $\lambda_{\min}(A)$ and $\lambda_{\max}(A)$ denote its smallest and largest eigenvalues.

The sets of $d \times d$ symmetric positive definite and symmetric positive semidefinite matrices are denoted by $\mathbb{S}^d_{++}$ and $\mathbb{S}^d_{+}$, respectively. For symmetric matrices of the same dimension, $A \preceq B$ means that $B - A$ is positive semidefinite, and $A \succ 0$ means that A is positive definite.

The operators $\mathbb{E}$, Var, and Cov denote population expectation, covariance, and cross-covariance, respectively, with Var reducing to variance for scalar arguments. Under local misspecification, $\mathbb{E}_n$ and Var_n denote expectation and covariance under the row-n distribution.

We write $\xrightarrow{\mathrm{p}}$ and $\xrightarrow{\mathrm{d}}$ for convergence in probability and distribution, respectively, and use $O_p(\cdot)$, $o_p(\cdot)$, and $\Theta_p(\cdot)$ in their standard senses. For nonnegative deterministic sequences, $a_n \lesssim b_n$ means that $a_n \leq C b_n$ for some constant $C > 0$ independent of n and all sufficiently large n. We write $a_n \asymp b_n$ when both $a_n \lesssim b_n$ and $b_n \lesssim a_n$.

B. Single-source deconfounding and causal interpretation

B.1 Additional details on the projected IV formulation

This subsection provides additional discussion of the projected IV formulation used in Section 2.2, including its relation to the classical IV moment, an explicit first-stage representation,

and the simplification of the first-order theory under valid instruments.

Dimension and identification of the projected IV moment. Recall that $Q_{\text{IV}} = \Sigma_{xz}\Sigma_z^{-1}\Sigma_{zx}$. When $\Sigma_z \in \mathbb{S}^q_{++}$, $\operatorname{rank}(Q_{\text{IV}}) = \operatorname{rank}(\Sigma_{zx}) \leq \min(p, q)$. Hence, identification of the full p-dimensional coefficient vector from the IV block alone requires $\operatorname{rank}(Q_{\text{IV}}) = p$, and therefore necessarily $q \geq p$.

When $q < p$, the projected IV moment $\mathbb{E}[(\Pi_z x_i)(y_i - x_i^\top \theta)]$ is still written as a p-dimensional vector, but its Jacobian has rank at most $q < p$. Thus, the p displayed equations contain fewer than p independent identifying directions and cannot uniquely identify the full coefficient vector.

The distinction between the raw and projected IV formulations is also relevant when $q \geq p$. The classical IV restriction $\mathbb{E}[z_i(y_i - x_i^\top \theta)] = 0_q$ contains q moment equations and, under $\operatorname{rank}(\Sigma_{zx}) = p$, is just identified when $q = p$ and overidentified when $q > p$. By contrast, $\mathbb{E}[(\Pi_z x_i)(y_i - x_i^\top \theta)] = \Sigma_{xz}\Sigma_z^{-1}\mathbb{E}[z_i(y_i - x_i^\top \theta)]$ is always p-dimensional. Under valid IV, $\mathbb{E}(z_i \varsigma_i) = 0_q$, the raw and projected moments identify the same structural coefficient under the corresponding rank condition. When $q > p$, however, projection may discard overidentifying directions contained in the raw IV moment. Consequently, efficiency results for the projected IV system do not necessarily extend to GMM estimators based on all raw IV moments.

Relation to an explicit first-stage representation. Some formulations of instrumental variables additionally posit $x_i = \eta^\top z_i + \epsilon_i$ with $\mathbb{E}(z_i \epsilon_i^\top) = 0$. Such a representation is useful when a structural interpretation of the first-stage regression is desired, but it is not

required for the projected-moment formulation used here.

Indeed, whenever $\Sigma_z \in \mathbb{S}^q_{++}$, the population linear projection is defined directly as $\Pi_z x_i = \Sigma_{xz}\Sigma_z^{-1} z_i$. Thus, the projected-IV statistical target and the corresponding 2SLS estimating equation depend only on this projection, the projected moment bias $\delta_{\text{IV}} := \mathbb{E}[(\Pi_z x_i)\varsigma_i]$, and the relevance matrix $Q_{\text{IV}} := \mathbb{E}[(\Pi_z x_i)x_i^\top]$. Accordingly, an explicit decomposition $x_i = \eta^\top z_i + \epsilon_i$ may be viewed as an optional linear-projection representation rather than a primitive assumption.

First-order representation under valid IV. Under the general projected-IV target, the first-order term is $\psi_{\text{IV},i} = (\Pi_z x_i)e_{\text{IV},i} + x_i z_i^\top \Sigma_z^{-1}\mathbb{E}[z_i(y_i - x_i^\top \beta_{\text{IV}})]$. The condition $\mathbb{E}[\|x_i z_i^\top\|_{\text{op}}^2] < \infty$ controls the second moment of the second component, whereas $\mathbb{E}\|z_i e_{\text{IV},i}\|_2^2 < \infty$ controls the projected-score component because $\Pi_z x_i = \Sigma_{xz}\Sigma_z^{-1} z_i$.

Under valid IV, $\mathbb{E}(z_i\varsigma_i) = 0_q$, we have $\delta_{\text{IV}} = 0_p$, $\beta_{\text{IV}} = \beta$, and $\mathbb{E}[z_i(y_i - x_i^\top \beta_{\text{IV}})] = 0_q$. Hence, the correction term vanishes and $e_{\text{IV},i} = \varsigma_i$, so that $\psi_{\text{IV},i} = (\Pi_z x_i)\varsigma_i$. Therefore, $\mathbb{E}[\|x_i z_i^\top\|_{\text{op}}^2] < \infty$ is unnecessary for asymptotic normality in the valid-IV case; it suffices to impose the corresponding finite-second-moment condition on $z_i\varsigma_i$, together with nondegeneracy of the limiting covariance matrix.

B.2 Additional details on surrogate-control residualization

This subsection provides further interpretation of the surrogate-control formulation in Section 2.3, with emphasis on the geometry of residualization, its projection representation, and the simplification of the first-order theory under exact surrogate-control validity.

Residualized relevance and projection geometry. Recall that $\Pi_w^{\perp} x_i = x_i - \Sigma_{xw}\Sigma_w^{-1} w_i$ and $Q_{\text{SV}} = \mathbb{E}[(\Pi_w^{\perp} x_i) x_i^{\top}]$. Since the population residual $\Pi_w^{\perp} x_i$ is orthogonal to w_i, we have

$$Q_{\text{SV}} = \mathbb{E}[(\Pi_w^{\perp} x_i)(\Pi_w^{\perp} x_i)^{\top}].$$

Hence, $Q_{\text{SV}} \in \mathbb{S}_{++}^{p}$ requires the residualized regressor to retain variation in every direction of the p-dimensional coefficient space.

This differs geometrically from IV relevance. The IV projection retains the component of x_i explained by z_i, whereas surrogate-control residualization removes the component of x_i explained by w_i. Thus, IV identification requires the projected component to span enough directions, whereas SV identification requires sufficiently rich variation to remain after partialling out the surrogate controls. This distinction motivates combining the two sources in the stacked moment system developed in the main text.

Partialling out as a projection representation. The surrogate-control estimator does not require a structural equation describing how w_i generates x_i. Its population construction only requires the linear projection $\Pi_w x_i = \Sigma_{xw}\Sigma_w^{-1} w_i$ and the residual $\Pi_w^{\perp} x_i = x_i - \Pi_w x_i$.

Equivalently, writing $\Gamma_w := \Sigma_w^{-1}\Sigma_{wx}$, one has the population decomposition $x_i = \Gamma_w^{\top} w_i + \Pi_w^{\perp} x_i$, with $\mathbb{E}[(\Pi_w^{\perp} x_i) w_i^{\top}] = 0$. The statistical target is determined by the residualized moment $\mathbb{E}[(\Pi_w^{\perp} x_i)(y_i - x_i^{\top}\theta)]$, rather than by a structural model for the relationship between x_i and w_i.

In particular, w_i itself need not be orthogonal to the structural disturbance. Under exact surrogate-control validity, $\mathbb{E}[(\Pi_w^{\perp} x_i)\varsigma_i] = 0_p$, which is equivalent to

$$\mathbb{E}(x_i\varsigma_i) = \Sigma_{xw}\Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i).$$

Thus, the surrogate controls absorb the component of regressor–disturbance dependence represented linearly through w_i, leaving residualized variation orthogonal to the disturbance.

First-order representation under exact surrogate-control validity.

Under the general statistical target $\beta_{\text{SV}} = \beta + Q_{\text{SV}}^{-1}\delta_{\text{SV}}$, define $r_{\text{SV},i} := y_i - x_i^{\top}\beta_{\text{SV}}$ and

$$e_{\text{SV},i} := r_{\text{SV},i} - w_i^{\top}\Sigma_w^{-1}\mathbb{E}(w_i r_{\text{SV},i}).$$

Then $\mathbb{E}(w_i e_{\text{SV},i}) = 0_m$, and the first-order term is $\psi_{\text{SV},i} = (\Pi_w^{\perp} x_i)e_{\text{SV},i}$.

The simplification relative to the IV case follows from the exact annihilation property $P_W^{\perp} W = 0$. Across the sample, the projected component of $r_{\text{SV},i}$ lies in the column space of W and is therefore annihilated by $P_W^{\perp}$. By contrast, $P_Z Z = Z$, so the corresponding instrument-span component is retained in the IV case and generates an additional first-order correction term under projected moment bias.

Under exact surrogate-control validity, $\delta_{\text{SV}} = 0_p$, we have $\beta_{\text{SV}} = \beta$ and $r_{\text{SV},i} = \varsigma_i$, so

$$e_{\text{SV},i} = \varsigma_i - w_i^{\top}\Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i).$$

Exact SV validity does not generally imply $\mathbb{E}(w_i\varsigma_i) = 0_m$, and hence $e_{\text{SV},i}$ need not equal ς_i. The first-order term therefore remains $\psi_{\text{SV},i} = (\Pi_w^{\perp} x_i)e_{\text{SV},i}$.

However, because $r_{\mathrm{SV},i} = \varsigma_i$, the condition $\mathbb{E}\|x_i x_i^\top\|_{\mathrm{op}}^2 < \infty$, which is needed under the general target to control the contribution of $x_i^\top Q_{\mathrm{SV}}^{-1}\delta_{\mathrm{SV}}$, is unnecessary for asymptotic normality under exact surrogate-control validity. The remaining second-moment conditions continue to control $x_i\varsigma_i$, the interaction between x_i and w_i, and the residualized score.

B.3 Additional discussion on causal interpretation

The results in Sections 2.2 and 2.3 are stated in terms of statistical moment conditions and statistical targets. This subsection provides additional discussion of the structural assumptions under which the no-bias conditions $\delta_{\mathrm{IV}} = 0$ and $\delta_{\mathrm{SV}} = 0$ admit causal interpretations.

IV moments and causal direction. In the linear IV formulation, exogeneity is commonly represented by $\mathbb{E}(z_i\varsigma_i) = 0_q$, while the projected formulation used in this paper requires the corresponding projected no-bias condition $\delta_{\mathrm{IV}} := \mathbb{E}[(\Pi_z x_i)\varsigma_i] = 0_p$. These moment restrictions are statistical statements and do not, by themselves, encode a causal direction between z_i and x_i.

The causal structure matters because it determines whether such orthogonality is scientifically plausible. For example, the classical IV interpretation views z_i as generating variation in x_i while remaining unrelated to the latent determinants of the outcome. By contrast, if z_i is constructed from, or strongly driven by, the endogenous regressor in a data-generating mechanism resembling $\varsigma_i \to x_i \to z_i$, then z_i may inherit information about ς_i, making the no-bias restriction difficult to justify. In such cases, the projected estimator remains statistically

defined, but its target is $\beta_{\text{IV}} = \beta + Q_{\text{IV}}^{-1}\delta_{\text{IV}}$.

Heterogeneous causal effects under IV. A common structural interpretation of β additionally relies on a homogeneous linear response. If the potential outcome instead takes the form $y_i(x) = \alpha_i + \beta_i^\top x$, or is nonlinear in x, the IV coefficient generally need not represent a universal structural slope.

For binary treatment, under the usual additional conditions including monotonicity, IV identifies a local average treatment effect for compliers rather than a population-wide common effect (Imbens and Angrist, 1994). More generally, the linear IV coefficient summarizes causal responses to instrument-induced variation in x_i. Without additional assumptions, it does not characterize the full causal response function.

This distinction is separate from moment invalidity. Even when $\delta_{\text{IV}} = 0$, effect heterogeneity may alter the causal interpretation of the IV estimand; when $\delta_{\text{IV}} \neq 0$, the target additionally contains the shift $Q_{\text{IV}}^{-1}\delta_{\text{IV}}$ induced by violation of the projected moment restriction.

Surrogate controls and causal direction. For surrogate-control adjustment, the statistical no-bias condition is $\delta_{\text{SV}} := \mathbb{E}[(\Pi_w^\perp x_i)\varsigma_i] = 0_p$. A natural causal justification is that w_i is a pre-treatment proxy for the latent confounding structure and captures the component of x_i associated with the structural disturbance.

The temporal or causal role of w_i is therefore important. If the data-generating mechanism is instead closer to $\varsigma_i \to x_i \to w_i$, then w_i may contain variation generated by the endogenous

regressor itself. Partialling out such a variable need not remove only latent confounding and may invalidate the desired causal interpretation. The estimator nevertheless remains centered at the statistical target $\beta_{\mathrm{sv}} = \beta + Q_{\mathrm{sv}}^{-1}\delta_{\mathrm{sv}}$, with $Q_{\mathrm{sv}}^{-1}\delta_{\mathrm{sv}}$ representing the remaining departure from the structural coefficient.

Relation to auxiliary-variable and proximal approaches. The surrogate-control formulation considered here is related to auxiliary-variable and proxy-based approaches to latent confounding. In particular, the auxiliary-variable framework of Miao et al. (2023) accommodates variables satisfying suitable exclusion-type conditions and develops nonparametric identification through equivalence and completeness conditions. Both approaches use observed auxiliary variables without requiring direct recovery of the latent confounder.

The identification targets and mechanisms are nevertheless different. The framework of Miao et al. (2023) concerns nonparametric identification of potential-outcome distributions, whereas the present paper focuses on a linear structural coefficient and the projection, rank, first-order, and weighting structure generated by IV and surrogate-control moments.

The present formulation is also related in spirit to proximal causal inference (Miao et al., 2018, 2024). Proximal methods typically introduce treatment-inducing and outcome-inducing proxies and identify causal effects through conditional moment equations involving a bridge function. Their identification theory relies on conditions such as proxy validity, existence of an appropriate bridge function, and completeness.

Our framework instead uses the direct residualized moment $\mathbb{E}[(\Pi_w^{\perp} x_i)(y_i - x_i^{\top}\theta)] = 0_p$.

We therefore do not solve a nonparametric bridge equation or attempt to recover the latent confounder. The surrogate variable enters through linear projection geometry: the component of x_i explained by w_i is removed, and inference uses the remaining variation.

Auxiliary-variable and proximal methods establish nonparametric causal identification under their respective assumptions. The linear model considered here yields IV and SV projected moments for a common coefficient vector, allowing us to study their joint identification, asymptotic behavior, and weighting.

C. Proof

C.1 Proof of Theorem 1

Proof of Theorem 1. Recall that $\Pi_z x_i = \Sigma_{xz}\Sigma_z^{-1} z_i$, $Q_{\text{IV}} = \mathbb{E}[(\Pi_z x_i)x_i^\top] = \Sigma_{xz}\Sigma_z^{-1}\Sigma_{zx}$, $\delta_{\text{IV}} = \mathbb{E}[(\Pi_z x_i)\varsigma_i]$, and $\beta_{\text{IV}} = \beta + Q_{\text{IV}}^{-1}\delta_{\text{IV}}$.

(i) Identification. Recall that the projected IV population moment condition is $\mathbb{E}[(\Pi_z x_i)(y_i - x_i^\top\theta)] = 0_p$, where $\theta \in \mathbb{R}^p$ is a candidate coefficient vector. Using $y_i = x_i^\top\beta + \varsigma_i$, we have

$$\mathbb{E}\Big[(\Pi_z x_i)\big(y_i - x_i^\top\theta\big)\Big] = \mathbb{E}\Big[(\Pi_z x_i)x_i^\top\Big](\beta - \theta) + \mathbb{E}[(\Pi_z x_i)\varsigma_i]\,.$$

By the definitions $Q_{\text{IV}} := \mathbb{E}[(\Pi_z x_i)x_i^\top]$ and $\delta_{\text{IV}} := \mathbb{E}[(\Pi_z x_i)\varsigma_i]$, the population moment condition is therefore equivalent to

$$Q_{\text{IV}}(\theta - \beta) = \delta_{\text{IV}}. \tag{C.1}$$

By Assumption 1(i), $Q_{\text{IV}} \in \mathbb{S}^p_{++}$, and hence Q_{IV} is nonsingular. Left-multiplying (C.1) by

Q_{IV}^{-1} gives

$$\theta = \beta + Q_{\text{IV}}^{-1}\delta_{\text{IV}} = \beta_{\text{IV}}.$$

Thus a solution exists.

To see that the solution is unique, suppose that both θ_1 and θ_2 satisfy (C.1). Then $Q_{\text{IV}}(\theta_1 - \theta_2) = 0_p$. Since Q_{IV} is nonsingular, its null space contains only 0_p, so $\theta_1 - \theta_2 = 0_p$, and hence $\theta_1 = \theta_2$. Therefore, the projected IV population moment uniquely identifies β_{IV}.

Finally, if $\delta_{\text{IV}} = 0_p$, then $Q_{\text{IV}}(\beta_{\text{IV}} - \beta) = 0_p$. Nonsingularity of Q_{IV} implies $\beta_{\text{IV}} = \beta$. Hence, when the projected IV moment bias vanishes, the projected IV population moment identifies the structural coefficient β.

Define $r_{\text{IV},i} := y_i - x_i^\top \beta_{\text{IV}} = \varsigma_i - x_i^\top Q_{\text{IV}}^{-1}\delta_{\text{IV}}$, and write $r_{\text{IV}} := (r_{\text{IV},1}, \ldots, r_{\text{IV},n})^\top = Y - X\beta_{\text{IV}}$. Then

$$\mathbb{E}[(\Pi_z x_i) r_{\text{IV},i}] = \delta_{\text{IV}} - Q_{\text{IV}} Q_{\text{IV}}^{-1}\delta_{\text{IV}} = 0_p.$$

Hence, β_{IV} satisfies the population projected normal equation

$$\Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i r_{\text{IV},i}) = 0_p. \tag{C.2}$$

Importantly, (C.2) does not generally imply $\mathbb{E}(z_i r_{\text{IV},i}) = 0_q$.

Using $Y = X\beta_{\text{IV}} + r_{\text{IV}}$, the 2SLS estimator satisfies the exact decomposition

$$\widehat{\beta}_{\text{IV}} - \beta_{\text{IV}} = \left(\frac{1}{n} X^\top P_Z X\right)^{-1} \left(\frac{1}{n} X^\top P_Z r_{\text{IV}}\right). \tag{C.3}$$

(ii) Consistency. Suppose Assumption 1 (i) holds. We first consider the denominator in (C.3).

Under the maintained second-moment conditions, the Cauchy–Schwarz inequality gives $\mathbb{E}\|x_i z_i^\top\|_{\mathrm{F}} \leq \{\mathbb{E}\|x_i\|_2^2\}^{1/2}\{\mathbb{E}\|z_i\|_2^2\}^{1/2} < \infty$, while $\mathbb{E}\|z_i z_i^\top\|_{\mathrm{F}} = \mathbb{E}\|z_i\|_2^2 < \infty$. Therefore, the weak law of large numbers applied entrywise gives $\frac{1}{n}X^\top Z \xrightarrow{\mathrm{p}} \Sigma_{xz}$, $\frac{1}{n}Z^\top Z \xrightarrow{\mathrm{p}} \Sigma_z$, and $\frac{1}{n}Z^\top X \xrightarrow{\mathrm{p}} \Sigma_{zx}$. Since $\Sigma_z \in \mathbb{S}_{++}^q$, the continuous mapping theorem gives $(\frac{1}{n}Z^\top Z)^{-1} \xrightarrow{\mathrm{p}} \Sigma_z^{-1}$. Thus,

$$\frac{1}{n}X^\top P_Z X = \left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1}\left(\frac{1}{n}Z^\top X\right) \xrightarrow{\mathrm{p}} Q_{\mathrm{IV}}. \tag{C.4}$$

Because $Q_{\mathrm{IV}} \in \mathbb{S}_{++}^p$, $(\frac{1}{n}X^\top P_Z X)^{-1} \xrightarrow{\mathrm{p}} Q_{\mathrm{IV}}^{-1}$.

We next consider the numerator. Since $r_{\mathrm{IV},i} = \varsigma_i - x_i^\top Q_{\mathrm{IV}}^{-1}\delta_{\mathrm{IV}}$, we have $\mathbb{E}(r_{\mathrm{IV},i}^2) \leq 2\mathbb{E}(\varsigma_i^2) + 2\|Q_{\mathrm{IV}}^{-1}\delta_{\mathrm{IV}}\|_2^2\mathbb{E}\|x_i\|_2^2 < \infty$. It follows from Cauchy–Schwarz that $\mathbb{E}\|z_i r_{\mathrm{IV},i}\|_2 < \infty$, and hence $\frac{1}{n}\sum_{i=1}^n z_i r_{\mathrm{IV},i} \xrightarrow{\mathrm{p}} \mathbb{E}(z_i r_{\mathrm{IV},i})$. Therefore,

$$\frac{1}{n}X^\top P_Z r_{\mathrm{IV}} = \left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1}\frac{1}{n}\sum_{i=1}^n z_i r_{\mathrm{IV},i} \xrightarrow{\mathrm{p}} \Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i r_{\mathrm{IV},i}) = 0_p,$$

where the last equality follows from (C.2).

Combining this result with (C.4) and (C.3) gives $\widehat{\beta}_{\mathrm{IV}} - \beta_{\mathrm{IV}} \xrightarrow{\mathrm{p}} 0_p$. Hence, $\widehat{\beta}_{\mathrm{IV}} \xrightarrow{\mathrm{p}} \beta_{\mathrm{IV}}$.

(iii) Asymptotic normality. Suppose further that Assumption 1 (ii) holds. Multiplying (C.3) by $\sqrt{n}$ gives

$$\sqrt{n}(\widehat{\beta}_{\mathrm{IV}} - \beta_{\mathrm{IV}}) = \left(\frac{1}{n}X^\top P_Z X\right)^{-1}\left(\frac{1}{\sqrt{n}}X^\top P_Z r_{\mathrm{IV}}\right). \tag{C.5}$$

By (C.4), the first factor converges in probability to Q_{IV}^{-1}. It therefore remains to derive the limiting distribution of the normalized numerator.

Directly expanding the numerator gives

$$\frac{1}{\sqrt{n}}X^\top P_Z r_{\text{IV}} = \left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^{n} z_i r_{\text{IV},i}. \tag{C.6}$$

The projected normal equation (C.2) only requires $\Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i r_{\text{IV},i}) = 0_p$, while $\mathbb{E}(z_i r_{\text{IV},i})$ itself need not vanish. Consequently, $n^{-1/2}\sum_{i=1}^n z_i r_{\text{IV},i}$ may contain a mean component of order $\sqrt{n}$, so this sum is not generally $O_p(1)$. Because this sum need not be $O_p(1)$, consistency of the estimated projection coefficient alone does not justify replacing it by its probability limit in (C.6).

To obtain a centered term, define

$$e_{\text{IV},i} := r_{\text{IV},i} - z_i^\top \Sigma_z^{-1}\mathbb{E}(z_i r_{\text{IV},i}).$$

The consistency argument gives $\mathbb{E}\|z_i r_{\text{IV},i}\|_2 < \infty$, so $\mathbb{E}(z_i r_{\text{IV},i})$ is finite. By construction,

$$\mathbb{E}(z_i e_{\text{IV},i}) = \mathbb{E}(z_i r_{\text{IV},i}) - \Sigma_z\Sigma_z^{-1}\mathbb{E}(z_i r_{\text{IV},i}) = 0_q.$$

Assumption 1 (ii) further gives $\mathbb{E}\|z_i e_{\text{IV},i}\|_2^2 < \infty$. Hence, the multivariate central limit theorem implies $n^{-1/2}\sum_{i=1}^n z_i e_{\text{IV},i} = O_p(1)$.

Writing $e_{\text{IV}} := (e_{\text{IV},1}, \ldots, e_{\text{IV},n})^\top$, the preceding definition gives $r_{\text{IV}} = e_{\text{IV}} + Z\Sigma_z^{-1}\mathbb{E}(z_i r_{\text{IV},i})$. At this point, the IV argument differs from the corresponding SV argument. For SV, the residual-maker eliminates the ancillary-variable component because $P_W^\perp W = 0$. For IV, however, $P_Z Z = Z$, so the component in the span of Z is retained. Substituting the decomposition of r_{IV} into (C.6) therefore gives the exact decomposition

$$\frac{1}{\sqrt{n}}X^\top P_Z r_{\text{IV}} = \left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^{n} z_i e_{\text{IV},i}$$

$$+ \frac{1}{\sqrt{n}} \sum_{i=1}^{n} x_i z_i^\top \Sigma_z^{-1} \mathbb{E}(z_i r_{\text{IV},i}). \tag{C.7}$$

Thus, in contrast to SV, recentering does not remove the ancillary-variable component but instead produces an additional first-order correction term.

For the first term in (C.7), the consistency argument gives

$$\left(\frac{1}{n} X^\top Z\right) \left(\frac{1}{n} Z^\top Z\right)^{-1} = \Sigma_{xz} \Sigma_z^{-1} + o_p(1).$$

Since $n^{-1/2} \sum_{i=1}^{n} z_i e_{\text{IV},i} = O_p(1)$, the estimation error in this coefficient contributes only $o_p(1)$. Hence,

$$\left(\frac{1}{n} X^\top Z\right) \left(\frac{1}{n} Z^\top Z\right)^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^{n} z_i e_{\text{IV},i} = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} (\Pi_z x_i) e_{\text{IV},i} + o_p(1).$$

Substituting this expression into (C.7) motivates defining

$$\psi_{\text{IV},i} := (\Pi_z x_i) e_{\text{IV},i} + x_i z_i^\top \Sigma_z^{-1} \mathbb{E}(z_i r_{\text{IV},i}). \tag{C.8}$$

The normalized numerator then admits the first-order representation

$$\frac{1}{\sqrt{n}} X^\top P_Z r_{\text{IV}} = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \psi_{\text{IV},i} + o_p(1). \tag{C.9}$$

It remains to verify that the summands in (C.9) are centered and have finite second moments. Using $\mathbb{E}(z_i e_{\text{IV},i}) = 0_q$,

$$\mathbb{E}(\psi_{\text{IV},i}) = \Sigma_{xz} \Sigma_z^{-1} \mathbb{E}(z_i e_{\text{IV},i}) + \Sigma_{xz} \Sigma_z^{-1} \mathbb{E}(z_i r_{\text{IV},i}) = 0_p,$$

where the last equality follows from (C.2).

For the second moment, $\mathbb{E}\|(\Pi_z x_i)e_{\text{IV},i}\|_2^2 \le \|\Sigma_{xz}\Sigma_z^{-1}\|_{\text{op}}^2 \mathbb{E}\|z_i e_{\text{IV},i}\|_2^2 < \infty$. Moreover,

$$\mathbb{E}\left\|x_i z_i^\top \Sigma_z^{-1}\mathbb{E}(z_i r_{\text{IV},i})\right\|_2^2 \le \left\|\Sigma_z^{-1}\mathbb{E}(z_i r_{\text{IV},i})\right\|_2^2 \mathbb{E}\|x_i z_i^\top\|_{\text{op}}^2 < \infty.$$

Therefore, $\mathbb{E}\|\psi_{\text{IV},i}\|_2^2 < \infty$. The second bound is precisely where the stronger condition $\mathbb{E}\|x_i z_i^\top\|_{\text{op}}^2 < \infty$ in Assumption 1(ii) is needed. Under valid IV, $\mathbb{E}(z_i r_{\text{IV},i}) = 0_q$, so the correction term in (C.8) vanishes and this additional bound is unnecessary for that term.

Because the observations are i.i.d., the $\psi_{\text{IV},i}$ are i.i.d., centered, and have finite second moments. Therefore, the multivariate central limit theorem gives

$$\frac{1}{\sqrt{n}}\sum_{i=1}^n \psi_{\text{IV},i} \xrightarrow{\text{d}} \mathcal{N}(0, \Sigma_{\text{IV}}), \qquad \Sigma_{\text{IV}} := \text{Var}(\psi_{\text{IV},i}). \tag{C.10}$$

Combining (C.9) and (C.10) yields $\frac{1}{\sqrt{n}} X^\top P_Z r_{\text{IV}} \xrightarrow{\text{d}} \mathcal{N}(0, \Sigma_{\text{IV}})$.

Finally, combining this result with (C.5), (C.4), and Slutsky's theorem gives

$$\sqrt{n}(\widehat{\beta}_{\text{IV}} - \beta_{\text{IV}}) \xrightarrow{\text{d}} \mathcal{N}\left(0, Q_{\text{IV}}^{-1}\Sigma_{\text{IV}}Q_{\text{IV}}^{-1}\right) = \mathcal{N}(0, V_{\text{IV}}), \tag{C.11}$$

where $V_{\text{IV}} := Q_{\text{IV}}^{-1}\Sigma_{\text{IV}}Q_{\text{IV}}^{-1}$. This completes the proof.

□

C.2 Proof of Theorem 2

Proof of Theorem 2. Recall that $\Pi_w^\perp x_i = x_i - \Sigma_{xw}\Sigma_w^{-1} w_i$, $Q_{\text{SV}} = \mathbb{E}\{(\Pi_w^\perp x_i)x_i^\top\}$, $\delta_{\text{SV}} = \mathbb{E}\{(\Pi_w^\perp x_i)\varsigma_i\}$, and $\beta_{\text{SV}} = \beta + Q_{\text{SV}}^{-1}\delta_{\text{SV}}$.

(i) Identification. Recall that the residualized population moment condition is $\mathbb{E}[(\Pi_w^{\perp} x_i)(y_i - x_i^{\top}\theta)] = 0_p$, where $\theta \in \mathbb{R}^p$ is a candidate coefficient vector. Using $y_i = x_i^{\top}\beta + \varsigma_i$, we have

$$\mathbb{E}\Big[(\Pi_w^{\perp} x_i)\big(y_i - x_i^{\top}\theta\big)\Big] = \mathbb{E}\Big[(\Pi_w^{\perp} x_i)x_i^{\top}\Big](\beta - \theta) + \mathbb{E}\Big[(\Pi_w^{\perp} x_i)\varsigma_i\Big].$$

By the definitions $Q_{\mathrm{SV}} := \mathbb{E}[(\Pi_w^{\perp} x_i)x_i^{\top}]$ and $\delta_{\mathrm{SV}} := \mathbb{E}[(\Pi_w^{\perp} x_i)\varsigma_i]$, the population moment condition is therefore equivalent to

$$Q_{\mathrm{SV}}(\theta - \beta) = \delta_{\mathrm{SV}}. \tag{C.12}$$

By Assumption 2(i), $Q_{\mathrm{SV}} \in \mathbb{S}_{++}^p$, and hence Q_{SV} is nonsingular. Left-multiplying (C.12) by Q_{SV}^{-1} gives

$$\theta = \beta + Q_{\mathrm{SV}}^{-1}\delta_{\mathrm{SV}} = \beta_{\mathrm{SV}}.$$

Thus a solution exists.

To see that the solution is unique, suppose that both θ_1 and θ_2 satisfy (C.12). Then $Q_{\mathrm{SV}}(\theta_1 - \theta_2) = 0_p$. Since Q_{SV} is nonsingular, its null space contains only 0_p, so $\theta_1 - \theta_2 = 0_p$, and hence $\theta_1 = \theta_2$. Therefore, the residualized population moment uniquely identifies β_{SV}.

Finally, if $\delta_{\mathrm{SV}} = 0_p$, then $Q_{\mathrm{SV}}(\beta_{\mathrm{SV}} - \beta) = 0_p$. Nonsingularity of Q_{SV} implies $\beta_{\mathrm{SV}} = \beta$. Hence, when the residualized moment bias vanishes, the surrogate-control population moment identifies the structural coefficient β.

Within the remainder of the proof, define $\Gamma_w := \Sigma_w^{-1}\Sigma_{wx}$, $\widetilde{x}_i := x_i - \Gamma_w^{\top} w_i = \Pi_w^{\perp} x_i$, and let $\widetilde{X}$ denote the matrix whose i-th row is $\widetilde{x}_i^{\top}$. Then

$$x_i = \Gamma_w^{\top} w_i + \widetilde{x}_i, \qquad X = W\Gamma_w + \widetilde{X}. \tag{C.13}$$

By the definition of the population linear projection,

$$\mathbb{E}(\widetilde{x}_i w_i^\top) = \Sigma_{xw} - \Gamma_w^\top \Sigma_w = 0_{p\times m}. \tag{C.14}$$

Consequently,

$$Q_{\text{SV}} = \mathbb{E}(\widetilde{x}_i x_i^\top) = \mathbb{E}(\widetilde{x}_i \widetilde{x}_i^\top) + \mathbb{E}(\widetilde{x}_i w_i^\top)\Gamma_w = \mathbb{E}(\widetilde{x}_i \widetilde{x}_i^\top).$$

Hence Q_{SV} is symmetric, and it is positive definite by Assumption 2(i).

Define $r_{\text{SV},i} := y_i - x_i^\top \beta_{\text{SV}} = \varsigma_i - x_i^\top Q_{\text{SV}}^{-1}\delta_{\text{SV}}$, and write $r_{\text{SV}} := (r_{\text{SV},1}, \ldots, r_{\text{SV},n})^\top$. Using $\widetilde{x}_i = \Pi_w^\perp x_i$, we obtain

$$\begin{aligned}
\mathbb{E}(\widetilde{x}_i r_{\text{SV},i}) &= \mathbb{E}(\widetilde{x}_i \varsigma_i) - \mathbb{E}(\widetilde{x}_i x_i^\top) Q_{\text{SV}}^{-1}\delta_{\text{SV}} \\
&= \delta_{\text{SV}} - Q_{\text{SV}} Q_{\text{SV}}^{-1}\delta_{\text{SV}} \\
&= 0_p.
\end{aligned} \tag{C.15}$$

Thus, β_{SV} satisfies the population residualized normal equation.

Since $P_W^\perp W = 0$, decomposition (C.13) gives $P_W^\perp X = P_W^\perp \widetilde{X}$. Because $P_W^\perp$ is symmetric, $X^\top P_W^\perp X = \widetilde{X}^\top P_W^\perp \widetilde{X}$ and $X^\top P_W^\perp r_{\text{SV}} = \widetilde{X}^\top P_W^\perp r_{\text{SV}}$. The estimator therefore satisfies the exact decomposition

$$\widehat{\beta}_{\text{SV}} - \beta_{\text{SV}} = \left(\frac{1}{n}\widetilde{X}^\top P_W^\perp \widetilde{X}\right)^{-1} \left(\frac{1}{n}\widetilde{X}^\top P_W^\perp r_{\text{SV}}\right). \tag{C.16}$$

(ii) Consistency. Suppose Assumption 2 (i) holds. The maintained second-moment conditions and the weak law of large numbers give $\frac{1}{n}\widetilde{X}^\top \widetilde{X} \xrightarrow{\text{p}} \mathbb{E}(\widetilde{x}_i \widetilde{x}_i^\top) = Q_{\text{SV}}$, $\frac{1}{n}\widetilde{X}^\top W \xrightarrow{\text{p}}$

$\mathbb{E}(\widetilde{x}_i w_i^\top) = 0_{p\times m}$, and $\frac{1}{n}W^\top W \xrightarrow{\mathrm{p}} \Sigma_w$. Since $\Sigma_w \in \mathbb{S}^m_{++}$,

$$\frac{1}{n}\widetilde{X}^\top P_W^\perp \widetilde{X} = \frac{1}{n}\widetilde{X}^\top \widetilde{X} - \left(\frac{1}{n}\widetilde{X}^\top W\right)\left(\frac{1}{n}W^\top W\right)^{-1}\left(\frac{1}{n}W^\top \widetilde{X}\right)$$

$$\xrightarrow{\mathrm{p}} Q_{\mathrm{SV}}.$$

It follows that $\{\frac{1}{n}\widetilde{X}^\top P_W^\perp \widetilde{X}\}^{-1} \xrightarrow{\mathrm{p}} Q_{\mathrm{SV}}^{-1}$.

Similarly, the weak law of large numbers gives $\frac{1}{n}\sum_{i=1}^n \widetilde{x}_i r_{\mathrm{SV},i} \xrightarrow{\mathrm{p}} \mathbb{E}(\widetilde{x}_i r_{\mathrm{SV},i}) = 0_p$ by (C.15), and $\frac{1}{n}\sum_{i=1}^n w_i r_{\mathrm{SV},i} \xrightarrow{\mathrm{p}} \mathbb{E}(w_i r_{\mathrm{SV},i})$. Therefore,

$$\frac{1}{n}\widetilde{X}^\top P_W^\perp r_{\mathrm{SV}} = \frac{1}{n}\sum_{i=1}^n \widetilde{x}_i r_{\mathrm{SV},i} - \left(\frac{1}{n}\widetilde{X}^\top W\right)\left(\frac{1}{n}W^\top W\right)^{-1}\left(\frac{1}{n}\sum_{i=1}^n w_i r_{\mathrm{SV},i}\right)$$

$$\xrightarrow{\mathrm{p}} 0_p.$$

Combining these limits with (C.16) yields

$$\widehat{\beta}_{\mathrm{SV}} \xrightarrow{\mathrm{p}} \beta_{\mathrm{SV}}.$$

(iii) Asymptotic normality. Suppose further that Assumption 2 (ii) holds. Multiplying (C.16) by $\sqrt{n}$ gives

$$\sqrt{n}(\widehat{\beta}_{\mathrm{SV}} - \beta_{\mathrm{SV}}) = \left(\frac{1}{n}\widetilde{X}^\top P_W^\perp \widetilde{X}\right)^{-1}\frac{1}{\sqrt{n}}\widetilde{X}^\top P_W^\perp r_{\mathrm{SV}}.$$

The first factor converges in probability to Q_{SV}^{-1}, so it remains to derive the limiting distribution of the normalized numerator.

A direct expansion gives

$$\frac{1}{\sqrt{n}}\widetilde{X}^\top P_W^\perp r_{\mathrm{SV}} = \frac{1}{\sqrt{n}}\sum_{i=1}^n \widetilde{x}_i r_{\mathrm{SV},i} - \left(\frac{1}{n}\widetilde{X}^\top W\right)\left(\frac{1}{n}W^\top W\right)^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^n w_i r_{\mathrm{SV},i}. \tag{C.17}$$

Although $\mathbb{E}(\widetilde{x}_i r_{\mathrm{SV},i}) = 0_p$, this does not imply $\mathbb{E}(w_i r_{\mathrm{SV},i}) = 0_m$. Hence the last sum in (C.17) may contain a nonzero mean component of order $\sqrt{n}$, so the second term cannot yet be treated as negligible.

To center this sum, define

$$c_{\mathrm{SV}} := \Sigma_w^{-1}\mathbb{E}(w_i r_{\mathrm{SV},i}), \qquad e_{\mathrm{SV},i} := r_{\mathrm{SV},i} - w_i^\top c_{\mathrm{SV}}. \tag{C.18}$$

Then $\mathbb{E}(w_i e_{\mathrm{SV},i}) = \mathbb{E}(w_i r_{\mathrm{SV},i}) - \Sigma_w c_{\mathrm{SV}} = 0_m$. Writing $e_{\mathrm{SV}} := (e_{\mathrm{SV},1}, \ldots, e_{\mathrm{SV},n})^\top$, equation (C.18) gives $r_{\mathrm{SV}} = e_{\mathrm{SV}} + W c_{\mathrm{SV}}$. Since $P_W^\perp W = 0$, we have the exact identity

$$P_W^\perp r_{\mathrm{SV}} = P_W^\perp e_{\mathrm{SV}}.$$

The corrected numerator therefore satisfies

$$\frac{1}{\sqrt{n}}\widetilde{X}^\top P_W^\perp r_{\mathrm{SV}} = \frac{1}{\sqrt{n}}\sum_{i=1}^n \widetilde{x}_i e_{\mathrm{SV},i} - \left(\frac{1}{n}\widetilde{X}^\top W\right)\left(\frac{1}{n}W^\top W\right)^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^n w_i e_{\mathrm{SV},i}.$$

Assumption 2(ii) and $\mathbb{E}(w_i e_{\mathrm{SV},i}) = 0_m$ imply $n^{-1/2}\sum_{i=1}^n w_i e_{\mathrm{SV},i} = O_p(1)$ by the multivariate central limit theorem. Moreover, $\frac{1}{n}\widetilde{X}^\top W = o_p(1)$ and $\{\frac{1}{n}W^\top W\}^{-1} = O_p(1)$. Thus, the second term is $o_p(1)$.

Define $\psi_{\mathrm{SV},i} := \widetilde{x}_i e_{\mathrm{SV},i} = (\Pi_w^\perp x_i) e_{\mathrm{SV},i}$. The preceding argument gives the first-order representation

$$\frac{1}{\sqrt{n}}X^\top P_W^\perp r_{\mathrm{SV}} = \frac{1}{\sqrt{n}}\sum_{i=1}^n \psi_{\mathrm{SV},i} + o_p(1). \tag{C.19}$$

The score is centered because, by (C.14) and (C.15),

$$\mathbb{E}(\psi_{\mathrm{SV},i}) = \mathbb{E}(\widetilde{x}_i r_{\mathrm{SV},i}) - \mathbb{E}(\widetilde{x}_i w_i^\top)c_{\mathrm{SV}} = 0_p.$$

It remains to verify that $\mathbb{E}\|\psi_{\mathrm{SV},i}\|_2^2 < \infty$. Since $r_{\mathrm{SV},i} = \varsigma_i - x_i^\top Q_{\mathrm{SV}}^{-1}\delta_{\mathrm{SV}}$,

$$\mathbb{E}\|x_i r_{\mathrm{SV},i}\|_2^2 \le 2\mathbb{E}\|x_i\varsigma_i\|_2^2 + 2\|Q_{\mathrm{SV}}^{-1}\delta_{\mathrm{SV}}\|_2^2\mathbb{E}\|x_i x_i^\top\|_{\mathrm{op}}^2 < \infty.$$

The consistency argument and the nonsingularity of Σ_w imply that c_{SV} in (C.18) is finite. Hence,

$$\mathbb{E}\|x_i e_{\mathrm{SV},i}\|_2^2 \le 2\mathbb{E}\|x_i r_{\mathrm{SV},i}\|_2^2 + 2\|c_{\mathrm{SV}}\|_2^2\mathbb{E}\|x_i w_i^\top\|_{\mathrm{op}}^2 < \infty.$$

Finally, $\psi_{\mathrm{SV},i} = x_i e_{\mathrm{SV},i} - \Gamma_w^\top w_i e_{\mathrm{SV},i}$, so

$$\mathbb{E}\|\psi_{\mathrm{SV},i}\|_2^2 \le 2\mathbb{E}\|x_i e_{\mathrm{SV},i}\|_2^2 + 2\|\Gamma_w^\top\|_{\mathrm{op}}^2\mathbb{E}\|w_i e_{\mathrm{SV},i}\|_2^2 < \infty.$$

Because the observations are i.i.d., the random vectors $\psi_{\mathrm{SV},i}$ are i.i.d., centered, and have finite second moments. The multivariate central limit theorem therefore gives

$$\frac{1}{\sqrt{n}}\sum_{i=1}^n \psi_{\mathrm{SV},i} \xrightarrow{\mathrm{d}} \mathcal{N}(0, \Sigma_{\mathrm{SV}}), \qquad \Sigma_{\mathrm{SV}} := \mathrm{Var}(\psi_{\mathrm{SV},i}).$$

Combining this result with (C.19) yields

$$\frac{1}{\sqrt{n}} X^\top P_W^\perp r_{\mathrm{SV}} \xrightarrow{\mathrm{d}} \mathcal{N}(0, \Sigma_{\mathrm{SV}}).$$

Finally, Slutsky's theorem gives

$$\sqrt{n}(\widehat{\beta}_{\mathrm{SV}} - \beta_{\mathrm{SV}}) \xrightarrow{\mathrm{d}} \mathcal{N}\left(0, Q_{\mathrm{SV}}^{-1}\Sigma_{\mathrm{SV}}Q_{\mathrm{SV}}^{-1}\right).$$

Thus,

$$V_{\mathrm{SV}} = Q_{\mathrm{SV}}^{-1}\Sigma_{\mathrm{SV}}Q_{\mathrm{SV}}^{-1}.$$

This completes the proof.

□

C.3 Proof of Proposition 3

Proof of Proposition 3. Recall that $h_i = \{(\Pi_z x_i)^\top, (\Pi_w^\perp x_i)^\top\}^\top$ and $J = \mathbb{E}(h_i x_i^\top) = \{Q_{\text{IV}}^\top, Q_{\text{SV}}^\top\}^\top$. Under exact projected validity, $\mathbb{E}(h_i \varsigma_i) = 0_{2p}$.

Population moment equation. Let $\theta \in \mathbb{R}^p$ be a candidate coefficient vector. Using $y_i = x_i^\top \beta + \varsigma_i$, the stacked population moment satisfies

$$\mathbb{E}\{h_i(y_i - x_i^\top \theta)\} = J(\beta - \theta) + \mathbb{E}(h_i \varsigma_i) = J(\beta - \theta).$$

Hence, θ satisfies the stacked moment restriction if and only if $J(\theta - \beta) = 0_{2p}$. Therefore, β is the unique solution if and only if $\ker(J) = \{0_p\}$, which is equivalent to $\text{rank}(J) = p$.

Rank characterization. Because J stacks the two projected Jacobian blocks, $Jv = 0_{2p}$ for $v \in \mathbb{R}^p$ if and only if $Q_{\text{IV}} v = 0_p$ and $Q_{\text{SV}} v = 0_p$. It follows that

$$\ker(J) = \ker(Q_{\text{IV}}) \cap \ker(Q_{\text{SV}}).$$

Consequently, $\text{rank}(J) = p$ if and only if $\ker(Q_{\text{IV}}) \cap \ker(Q_{\text{SV}}) = \{0_p\}$, which proves the results.

Complementarity. The preceding characterization does not require either $\ker(Q_{\text{IV}})$ or $\ker(Q_{\text{SV}})$ to be trivial individually. Thus, either projected moment block may be rank deficient as long as no nonzero parameter direction belongs to both null spaces. Equivalently, directions that are not identified by one block may still be identified by the other, and the stacked system uniquely identifies β whenever their unidentified directions have trivial intersection.

□

C.4 Proof of Theorem 4

Proof of Theorem 4. Recall that $\varsigma_i = y_i - x_i^\top \beta$ and

$$\widehat{g}(\theta) = \frac{1}{n} \begin{pmatrix} X^\top P_Z (Y - X\theta) \\ \\ X^\top P_W^\perp (Y - X\theta) \end{pmatrix}.$$

Recall that

$$\widehat{J} = \frac{1}{n} \begin{pmatrix} X^\top P_Z X \\ \\ X^\top P_W^\perp X \end{pmatrix}.$$

Since the sample moment is linear in θ,

$$\widehat{g}(\theta) = \widehat{g}(\beta) - \widehat{J}(\theta - \beta). \tag{C.20}$$

Assumption 3 (i) gives $\mathbb{E}\{(\Pi_z x_i)\varsigma_i\} = 0_p$ and $\mathbb{E}\{(\Pi_w^\perp x_i)\varsigma_i\} = 0_p$. Thus, both projected moment blocks are correctly specified at β.

Identification. Let $g(\theta)$ denote the population counterpart of $\widehat{g}(\theta)$. By linearity and the projected moment restrictions,

$$g(\theta) = -J(\theta - \beta).$$

Hence $g(\theta) = 0_{2p}$ if and only if $J(\theta - \beta) = 0_{2p}$. Since Assumption 3(i) gives $\operatorname{rank}(J) = p$, the null space of J is trivial. Therefore, $\theta = \beta$, so β is the unique solution to the population combined projected moment condition.

The first-order condition for $\widehat{\beta}(M)$ is $\widehat{J}^\top M \widehat{g}\{\widehat{\beta}(M)\} = 0_p$. Using (C.20), we obtain the exact estimation-error decomposition

$$\widehat{\beta}(M) - \beta = \left(\widehat{J}^\top M \widehat{J}\right)^{-1} \widehat{J}^\top M \widehat{g}(\beta). \tag{C.21}$$

Consistency. Suppose Assumption 3 (i) holds. We first establish the probability limit of $\widehat{J}$.

For the IV block, the maintained second-moment conditions and the weak law of large numbers give $\frac{1}{n}X^\top Z \xrightarrow{\text{p}} \Sigma_{xz}$, $\frac{1}{n}Z^\top Z \xrightarrow{\text{p}} \Sigma_z$, and $\frac{1}{n}Z^\top X \xrightarrow{\text{p}} \Sigma_{zx}$. Since $\Sigma_z \in \mathbb{S}^q_{++}$, $(\frac{1}{n}Z^\top Z)^{-1} \xrightarrow{\text{p}} \Sigma_z^{-1}$. Hence,

$$\frac{1}{n}X^\top P_Z X = \left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1}\left(\frac{1}{n}Z^\top X\right) \xrightarrow{\text{p}} Q_{\text{IV}}.$$

For the SV block, the weak law of large numbers gives $\frac{1}{n}X^\top X \xrightarrow{\text{p}} \Sigma_x$, $\frac{1}{n}X^\top W \xrightarrow{\text{p}} \Sigma_{xw}$, $\frac{1}{n}W^\top W \xrightarrow{\text{p}} \Sigma_w$, and $\frac{1}{n}W^\top X \xrightarrow{\text{p}} \Sigma_{wx}$. Since $\Sigma_w \in \mathbb{S}^m_{++}$, $(\frac{1}{n}W^\top W)^{-1} \xrightarrow{\text{p}} \Sigma_w^{-1}$. Therefore,

$$\frac{1}{n}X^\top P_W^\perp X \xrightarrow{\text{p}} \Sigma_x - \Sigma_{xw}\Sigma_w^{-1}\Sigma_{wx} = Q_{\text{SV}}.$$

Combining the two blocks gives

$$\widehat{J} \xrightarrow{\text{p}} \begin{pmatrix} Q_{\text{IV}} \\ \\ Q_{\text{SV}} \end{pmatrix} = J. \tag{C.22}$$

We next establish the probability limit of $\widehat{g}(\beta)$. For the IV block,

$$\frac{1}{n}X^\top P_Z \varsigma = \left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1}\left(\frac{1}{n}\sum_{i=1}^n z_i\varsigma_i\right) \xrightarrow{\text{p}} \Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i) = 0_p,$$

where the final equality follows from $\mathbb{E}\{(\Pi_z x_i)\varsigma_i\} = 0_p$. In particular, this does not require the stronger raw IV restriction $\mathbb{E}(z_i\varsigma_i) = 0_q$.

For the SV block,

$$\frac{1}{n}X^\top P_W^\perp \varsigma \xrightarrow{\text{p}} \mathbb{E}(x_i\varsigma_i) - \Sigma_{xw}\Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i) = \mathbb{E}\{(\Pi_w^\perp x_i)\varsigma_i\} = 0_p.$$

Hence,

$$\widehat{g}(\beta) \xrightarrow{\mathrm{p}} 0_{2p}. \tag{C.23}$$

By Assumption 3 (i), $J^\top M J \in \mathbb{S}^p_{++}$. Together with (C.22), this implies that $\widehat{J}^\top M \widehat{J}$ is positive definite with probability approaching one. The continuous mapping theorem therefore gives

$$\left(\widehat{J}^\top M \widehat{J}\right)^{-1} \widehat{J}^\top M \xrightarrow{\mathrm{p}} (J^\top M J)^{-1} J^\top M = K_M.$$

Combining this result with (C.21) and (C.23) gives $\widehat{\beta}(M) - \beta \xrightarrow{\mathrm{p}} 0_p$. Therefore, $\widehat{\beta}(M) \xrightarrow{\mathrm{p}} \beta$.

Asymptotic normality. Suppose further that Assumption 3 (ii) holds. Multiplying (C.21) by $\sqrt{n}$ gives

$$\sqrt{n}\{\widehat{\beta}(M) - \beta\} = \left(\widehat{J}^\top M \widehat{J}\right)^{-1} \widehat{J}^\top M \sqrt{n}\, \widehat{g}(\beta). \tag{C.24}$$

By the consistency argument, the matrix multiplying $\sqrt{n}\,\widehat{g}(\beta)$ converges to K_M. It therefore remains to derive a central limit theorem for the two projected moment blocks in $\sqrt{n}\,\widehat{g}(\beta)$.

Define the population projection residuals

$$e_{z,i} := \varsigma_i - z_i^\top \Sigma_z^{-1} \mathbb{E}(z_i \varsigma_i), \qquad e_{w,i} := \varsigma_i - w_i^\top \Sigma_w^{-1} \mathbb{E}(w_i \varsigma_i).$$

The maintained second-moment conditions ensure that the disturbance cross moments are finite. By construction,

$$\mathbb{E}(z_i e_{z,i}) = 0_q, \qquad \mathbb{E}(w_i e_{w,i}) = 0_m.$$

The standing second-moment conditions imply that $\mathbb{E}(z_i \varsigma_i)$ and $\mathbb{E}(w_i \varsigma_i)$ are finite, so c_z and c_w are finite nonrandom vectors. By construction, $\mathbb{E}(z_i e_{z,i}) = 0_q$ and $\mathbb{E}(w_i e_{w,i}) = 0_m$.

We first consider the projected IV block. A direct expansion in terms of ς_i cannot generally be treated by replacing the sample projection coefficient by its probability limit, because the projected restriction $\Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i) = 0_p$ does not imply $\mathbb{E}(z_i\varsigma_i) = 0_q$. Thus, $n^{-1/2}\sum_{i=1}^n z_i\varsigma_i$ may contain a mean component of order $\sqrt{n}$.

Using $\varsigma_i = e_{z,i} + z_i^\top c_z$, the IV block admits the exact decomposition

$$\frac{1}{\sqrt{n}}X^\top P_Z\varsigma = \left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^n z_i e_{z,i} + \frac{1}{\sqrt{n}}\sum_{i=1}^n x_i z_i^\top c_z. \tag{C.25}$$

The second term remains because $P_Z Z = Z$; hence, the component of the structural error lying in the span of Z is retained by the IV projection.

Assumption 3 (ii) gives $\mathbb{E}\|z_i e_{z,i}\|_2^2 < \infty$, so $n^{-1/2}\sum_{i=1}^n z_i e_{z,i} = O_p(1)$. Moreover,

$$\left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1} = \Sigma_{xz}\Sigma_z^{-1} + o_p(1).$$

Therefore, the first term in (C.25) equals $n^{-1/2}\sum_{i=1}^n(\Pi_z x_i)e_{z,i} + o_p(1)$, and hence

$$\frac{1}{\sqrt{n}}X^\top P_Z\varsigma = \frac{1}{\sqrt{n}}\sum_{i=1}^n\left\{(\Pi_z x_i)e_{z,i} + x_i z_i^\top c_z\right\} + o_p(1). \tag{C.26}$$

We next consider the projected SV block. For the SV block, the projection decomposition leads to a different simplification. Since $\varsigma_i = e_{w,i} + w_i^\top c_w$ and $P_W^\perp W = 0$, the component lying in the span of W is removed exactly:

$$\frac{1}{\sqrt{n}}X^\top P_W^\perp\varsigma = \frac{1}{\sqrt{n}}X^\top P_W^\perp e_w,$$

where $e_w = (e_{w,1}, \ldots, e_{w,n})^\top$.

Let $\Pi_w^\perp X$ denote the matrix whose i-th row is $(\Pi_w^\perp x_i)^\top$. Since $X = \Pi_w^\perp X + W\Sigma_w^{-1}\Sigma_{wx}$, the identity $P_W^\perp W = 0$ implies $P_W^\perp X = P_W^\perp(\Pi_w^\perp X)$. Expanding the sample residual-maker therefore gives

$$\frac{1}{\sqrt{n}} X^\top P_W^\perp \varsigma = \frac{1}{\sqrt{n}} \sum_{i=1}^n (\Pi_w^\perp x_i) e_{w,i} - \left\{ \frac{1}{n} \sum_{i=1}^n (\Pi_w^\perp x_i) w_i^\top \right\} \left(\frac{1}{n} \sum_{i=1}^n w_i w_i^\top \right)^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n w_i e_{w,i}. \tag{C.27}$$

By the definition of the population residualization, $\mathbb{E}\{(\Pi_w^\perp x_i) w_i^\top\} = 0_{p \times m}$. The weak law of large numbers therefore gives $n^{-1} \sum_{i=1}^n (\Pi_w^\perp x_i) w_i^\top = o_p(1)$. Also, $(n^{-1} \sum_{i=1}^n w_i w_i^\top)^{-1} \xrightarrow{\mathrm{P}} \Sigma_w^{-1}$, while $\mathbb{E}(w_i e_{w,i}) = 0_m$ and $\mathbb{E}\|w_i e_{w,i}\|_2^2 < \infty$ imply $n^{-1/2} \sum_{i=1}^n w_i e_{w,i} = O_p(1)$. Hence, the second term in (C.27) is $o_p(1)$, so

$$\frac{1}{\sqrt{n}} X^\top P_W^\perp \varsigma = \frac{1}{\sqrt{n}} \sum_{i=1}^n (\Pi_w^\perp x_i) e_{w,i} + o_p(1). \tag{C.28}$$

Equations (C.26) and (C.28) collect the first-order variation of the two feasible projected moment blocks. Recall from Section 3.1 that $\psi_{\mathrm{IV},i}$ and $\psi_{\mathrm{SV},i}$ denote the corresponding first-order scores. Under the notation introduced above, these previously defined scores admit the equivalent representations

$$\psi_{\mathrm{IV},i} = (\Pi_z x_i) e_{z,i} + x_i z_i^\top c_z, \qquad \psi_{\mathrm{SV},i} = (\Pi_w^\perp x_i) e_{w,i}.$$

Recall also that $\psi_i = (\psi_{\mathrm{IV},i}^\top, \psi_{\mathrm{SV},i}^\top)^\top$ and $\Sigma = \mathrm{Var}(\psi_i)$. Stacking (C.26) and (C.28) gives the

first-order representation

$$\sqrt{n}\,\widehat{g}(\beta) = \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\psi_i + o_p(1). \tag{C.29}$$

It remains to verify that the summands in (C.29) are centered and have finite second moments.

For the IV component, using $\mathbb{E}(z_i e_{z,i}) = 0_q$,

$$\mathbb{E}(\psi_{\text{IV},i}) = \Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i e_{z,i}) + \Sigma_{xz}c_z = \Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i) = 0_p,$$

where the last equality follows from the projected IV restriction. For the SV component,

$$\mathbb{E}(\psi_{\text{SV},i}) = \mathbb{E}\{(\Pi_w^{\perp}x_i)\varsigma_i\} - \mathbb{E}\{(\Pi_w^{\perp}x_i)w_i^{\top}\}c_w = 0_p.$$

Hence,

$$\mathbb{E}(\psi_i) = \begin{pmatrix} \mathbb{E}(\psi_{\text{IV},i}) \\ \\ \mathbb{E}(\psi_{\text{SV},i}) \end{pmatrix} = \begin{pmatrix} 0_p \\ \\ 0_p \end{pmatrix} = 0_{2p}.$$

For the IV second moment, $\mathbb{E}\|(\Pi_z x_i)e_{z,i}\|_2^2 \le \|\Sigma_{xz}\Sigma_z^{-1}\|_{\text{op}}^2\mathbb{E}\|z_i e_{z,i}\|_2^2 < \infty$. Moreover, $\mathbb{E}\|x_i z_i^{\top}c_z\|_2^2 \le \|c_z\|_2^2\mathbb{E}\|x_i z_i^{\top}\|_{\text{op}}^2 < \infty$. Thus, $\mathbb{E}\|\psi_{\text{IV},i}\|_2^2 < \infty$.

For the SV component, first note that $x_i e_{w,i} = x_i\varsigma_i - x_i w_i^{\top}c_w$. Therefore,

$$\mathbb{E}\|x_i e_{w,i}\|_2^2 \le 2\mathbb{E}\|x_i\varsigma_i\|_2^2 + 2\|c_w\|_2^2\mathbb{E}\|x_i w_i^{\top}\|_{\text{op}}^2 < \infty.$$

Since $(\Pi_w^{\perp}x_i)e_{w,i} = x_i e_{w,i} - \Sigma_{xw}\Sigma_w^{-1}w_i e_{w,i}$, we further have

$$\mathbb{E}\|\psi_{\text{SV},i}\|_2^2 \le 2\mathbb{E}\|x_i e_{w,i}\|_2^2 + 2\|\Sigma_{xw}\Sigma_w^{-1}\|_{\text{op}}^2\mathbb{E}\|w_i e_{w,i}\|_2^2 < \infty.$$

Consequently, $\mathbb{E}\|\psi_i\|_2^2 < \infty$, so $\Sigma = \text{Var}(\psi_i)$ is well defined.

Because the observations are i.i.d., the ψ_i are i.i.d., centered, and have finite second moments. The multivariate central limit theorem therefore gives

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} \psi_i \xrightarrow{\mathrm{d}} \mathcal{N}(0, \Sigma). \tag{C.30}$$

Combining (C.29) and (C.30) yields

$$\sqrt{n}\, \widehat{g}(\beta) \xrightarrow{\mathrm{d}} \mathcal{N}(0, \Sigma).$$

Finally, by (C.22),

$$\left(\widehat{J}^\top M \widehat{J}\right)^{-1} \widehat{J}^\top M \xrightarrow{\mathrm{p}} (J^\top M J)^{-1} J^\top M = K_M.$$

Combining this result with (C.24) and Slutsky's theorem gives

$$\sqrt{n}\{\widehat{\beta}(M) - \beta\} \xrightarrow{\mathrm{d}} \mathcal{N}\left(0,\, K_M \Sigma K_M^\top\right) = \mathcal{N}\{0, V(M)\}, \tag{C.31}$$

where $V(M) = K_M \Sigma K_M^\top$. By Assumption 3 (ii), $V(M) \in \mathbb{S}^p_{++}$, so the limiting distribution is nondegenerate. This completes the proof.

□

C.5 Proof of Theorem 5

Proof of Theorem 5. Recall that

$$\widehat{g}(\theta) = \frac{1}{n} \begin{pmatrix} X^\top P_Z (Y - X\theta) \\ \\ X^\top P_W^\perp (Y - X\theta) \end{pmatrix}, \qquad \widehat{J} = \frac{1}{n} \begin{pmatrix} X^\top P_Z X \\ \\ X^\top P_W^\perp X \end{pmatrix}.$$

Since the sample moment is linear in θ,

$$\widehat{g}(\theta) = \widehat{g}(\beta) - \widehat{J}(\theta - \beta).$$

Throughout the local sequence, the joint distribution of (x_i, z_i, w_i) is fixed. Hence the population projections, design second moments, J, and K_M do not depend on n. Expectations involving ς_i are taken under the row-n distribution and denoted by $\mathbb{E}_n$.

Population target. Let $g_n(\theta) := \mathbb{E}_n\{h_i(y_i - x_i^\top \theta)\}$. Using $y_i = x_i^\top \beta + \varsigma_i$ and (3.6),

$$g_n(\theta) = \frac{\tau}{\sqrt{n}} - J(\theta - \beta).$$

The population GMM criterion is

$$g_n(\theta)^\top M g_n(\theta).$$

By Assumption 4 (i), $J^\top M J \in \mathbb{S}^p_{++}$, so the criterion is strictly convex. Its unique minimizer therefore satisfies

$$J^\top M \left\{ \frac{\tau}{\sqrt{n}} - J(\beta_{M,n} - \beta) \right\} = 0_p.$$

Hence

$$\beta_{M,n} - \beta = \frac{1}{\sqrt{n}} (J^\top M J)^{-1} J^\top M \tau = \frac{1}{\sqrt{n}} K_M \tau,$$

and therefore

$$\beta_{M,n} = \beta + n^{-1/2} K_M \tau \longrightarrow \beta.$$

Consistency. Suppose further that Assumption 4 (ii) holds. The observations are i.i.d. within each row, and the assumed second moments imply, by Chebyshev's inequality applied entrywise,

$$\frac{1}{n}X^\top Z = \Sigma_{xz} + O_p(n^{-1/2}), \qquad \frac{1}{n}Z^\top Z = \Sigma_z + O_p(n^{-1/2}),$$

$$\frac{1}{n}X^\top X = \Sigma_x + O_p(n^{-1/2}), \qquad \frac{1}{n}X^\top W = \Sigma_{xw} + O_p(n^{-1/2}),$$

$$\frac{1}{n}W^\top W = \Sigma_w + O_p(n^{-1/2}).$$

Since Σ_z and Σ_w are positive definite,

$$\left(\frac{1}{n}Z^\top Z\right)^{-1} \xrightarrow{\text{p}} \Sigma_z^{-1}, \qquad \left(\frac{1}{n}W^\top W\right)^{-1} \xrightarrow{\text{p}} \Sigma_w^{-1}.$$

It follows that

$$\frac{1}{n}X^\top P_Z X = \left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1}\left(\frac{1}{n}Z^\top X\right) \xrightarrow{\text{p}} Q_{\text{IV}},$$

and

$$\frac{1}{n}X^\top P_W^\perp X = \frac{1}{n}X^\top X - \left(\frac{1}{n}X^\top W\right)\left(\frac{1}{n}W^\top W\right)^{-1}\left(\frac{1}{n}W^\top X\right)$$

$$\xrightarrow{\text{p}} \Sigma_x - \Sigma_{xw}\Sigma_w^{-1}\Sigma_{wx} = Q_{\text{SV}}.$$

Therefore,

$$\widehat{J} \xrightarrow{\text{p}} \begin{pmatrix} Q_{\text{IV}} \\ \\ Q_{\text{SV}} \end{pmatrix} = J.$$

Since $J^\top M J \in \mathbb{S}^p_{++}$, the matrix $\widehat{J}^\top M \widehat{J}$ is nonsingular with probability approaching one, and

$$(\widehat{J}^\top M \widehat{J})^{-1}\widehat{J}^\top M \xrightarrow{\text{p}} (J^\top M J)^{-1} J^\top M = K_M. \tag{C.32}$$

It remains to control the sample moments at β. Define

$$e_{z,i} := \varsigma_i - z_i^\top \Sigma_z^{-1}\mathbb{E}_n(z_i\varsigma_i), \qquad e_{w,i} := \varsigma_i - w_i^\top \Sigma_w^{-1}\mathbb{E}_n(w_i\varsigma_i).$$

By construction,

$$\mathbb{E}_n(z_i e_{z,i}) = 0_q, \qquad \mathbb{E}_n(w_i e_{w,i}) = 0_m.$$

Cauchy–Schwarz and the fixed design second moments give

$$\sup_n \|\mathbb{E}_n(z_i\varsigma_i)\|_2 < \infty, \qquad \sup_n \|\mathbb{E}_n(w_i\varsigma_i)\|_2 < \infty.$$

Moreover,

$$z_i\varsigma_i = z_i e_{z,i} + z_i z_i^\top \Sigma_z^{-1}\mathbb{E}_n(z_i\varsigma_i),$$

so Assumption 4 (ii) implies

$$\sup_n \mathbb{E}_n\|z_i\varsigma_i\|_2^2 < \infty.$$

The same argument gives

$$\sup_n \mathbb{E}_n\|w_i\varsigma_i\|_2^2 < \infty.$$

Together with $\sup_n \mathbb{E}_n\|x_i\varsigma_i\|_2^2 < \infty$, Chebyshev's inequality yields

$$\frac{1}{n}X^\top\varsigma = \mathbb{E}_n(x_i\varsigma_i) + O_p(n^{-1/2}),$$

$$\frac{1}{n}Z^\top\varsigma = \mathbb{E}_n(z_i\varsigma_i) + O_p(n^{-1/2}),$$

$$\frac{1}{n}W^\top\varsigma = \mathbb{E}_n(w_i\varsigma_i) + O_p(n^{-1/2}).$$

Using

$$\frac{1}{n}X^\top P_Z\varsigma = \left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1}\left(\frac{1}{n}Z^\top\varsigma\right)$$

and

$$\frac{1}{n}X^\top P_W^\perp\varsigma = \frac{1}{n}X^\top\varsigma - \left(\frac{1}{n}X^\top W\right)\left(\frac{1}{n}W^\top W\right)^{-1}\left(\frac{1}{n}W^\top\varsigma\right),$$

we obtain

$$\widehat{g}(\beta) - g_n(\beta) = O_p(n^{-1/2}).$$

Since

$$g_n(\beta) = \frac{\tau}{\sqrt{n}},$$

it follows that

$$\widehat{g}(\beta) = O_p(n^{-1/2}) \xrightarrow{\mathrm{p}} 0_{2p}.$$

The first-order condition for $\widehat{\beta}(M)$, together with the linearity of $\widehat{g}(\theta)$, gives the exact decomposition

$$\widehat{\beta}(M) - \beta = (\widehat{J}^\top M \widehat{J})^{-1} \widehat{J}^\top M \widehat{g}(\beta). \tag{C.33}$$

Combining (C.32) with $\widehat{g}(\beta) \xrightarrow{\mathrm{p}} 0_{2p}$ gives

$$\widehat{\beta}(M) \xrightarrow{\mathrm{p}} \beta.$$

Asymptotic normality. Suppose further that Assumption 4 (iii) holds. Multiplying (C.33) by $\sqrt{n}$,

$$\sqrt{n}\{\widehat{\beta}(M) - \beta\} = (\widehat{J}^\top M \widehat{J})^{-1} \widehat{J}^\top M \sqrt{n}\, \widehat{g}(\beta).$$

By (C.32), the matrix multiplying $\sqrt{n}\, \widehat{g}(\beta)$ converges in probability to K_M. It therefore remains to derive the limiting distribution of $\sqrt{n}\, \widehat{g}(\beta)$.

For the IV block, the score in (3.4) can be written as

$$\psi_{\mathrm{IV},i} = \Sigma_{xz} \Sigma_z^{-1} z_i e_{z,i} + x_i z_i^\top \Sigma_z^{-1} \mathbb{E}_n(z_i \varsigma_i).$$

A direct calculation gives

$$\begin{aligned}&\frac{1}{\sqrt{n}}X^\top P_Z\varsigma - \frac{1}{\sqrt{n}}\sum_{i=1}^n \psi_{\mathrm{IV},i}\\&\quad= \left\{\left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1} - \Sigma_{xz}\Sigma_z^{-1}\right\}\frac{1}{\sqrt{n}}\sum_{i=1}^n z_i e_{z,i}.\end{aligned}$$

The matrix in braces is $o_p(1)$. Moreover,

$$\mathbb{E}_n(z_i e_{z,i}) = 0_q, \qquad \sup_n \mathbb{E}_n\|z_i e_{z,i}\|_2^2 < \infty,$$

so

$$\frac{1}{\sqrt{n}}\sum_{i=1}^n z_i e_{z,i} = O_p(1).$$

Hence

$$\frac{1}{\sqrt{n}}X^\top P_Z\varsigma = \frac{1}{\sqrt{n}}\sum_{i=1}^n \psi_{\mathrm{IV},i} + o_p(1).$$

For the SV block, $P_W^\perp W = 0$ gives

$$X^\top P_W^\perp \varsigma = X^\top P_W^\perp e_w,$$

where $e_w = (e_{w,1}, \ldots, e_{w,n})^\top$. Since

$$\psi_{\mathrm{SV},i} = (x_i - \Sigma_{xw}\Sigma_w^{-1}w_i)e_{w,i},$$

we similarly obtain

$$\begin{aligned}&\frac{1}{\sqrt{n}}X^\top P_W^\perp \varsigma - \frac{1}{\sqrt{n}}\sum_{i=1}^n \psi_{\mathrm{SV},i}\\&\quad= -\left\{\left(\frac{1}{n}X^\top W\right)\left(\frac{1}{n}W^\top W\right)^{-1} - \Sigma_{xw}\Sigma_w^{-1}\right\}\frac{1}{\sqrt{n}}\sum_{i=1}^n w_i e_{w,i}.\end{aligned}$$

Again,

$$\left(\frac{1}{n}X^\top W\right)\left(\frac{1}{n}W^\top W\right)^{-1} - \Sigma_{xw}\Sigma_w^{-1} = o_p(1),$$

while

$$\mathbb{E}_n(w_i e_{w,i}) = 0_m, \qquad \frac{1}{\sqrt{n}}\sum_{i=1}^n w_i e_{w,i} = O_p(1).$$

Therefore,

$$\frac{1}{\sqrt{n}}X^\top P_W^\perp \varsigma = \frac{1}{\sqrt{n}}\sum_{i=1}^n \psi_{\text{SV},i} + o_p(1).$$

Stacking the two blocks gives

$$\sqrt{n}\,\widehat{g}(\beta) = \frac{1}{\sqrt{n}}\sum_{i=1}^n \psi_i + o_p(1). \tag{C.34}$$

We next determine the mean of the score. Using

$$\mathbb{E}\{(x_i - \Pi_z x_i)z_i^\top\} = 0_{p\times q}, \qquad \mathbb{E}\{(\Pi_w^\perp x_i)w_i^\top\} = 0_{p\times m},$$

we have

$$\mathbb{E}_n(\psi_{\text{IV},i}) = \mathbb{E}_n\{(\Pi_z x_i)\varsigma_i\}, \qquad \mathbb{E}_n(\psi_{\text{SV},i}) = \mathbb{E}_n\{(\Pi_w^\perp x_i)\varsigma_i\}.$$

Hence, by (3.6),

$$\mathbb{E}_n(\psi_i) = \mathbb{E}_n(h_i\varsigma_i) = \frac{\tau}{\sqrt{n}}.$$

Consequently,

$$\frac{1}{\sqrt{n}}\sum_{i=1}^n \psi_i = \tau + \frac{1}{\sqrt{n}}\sum_{i=1}^n\{\psi_i - \mathbb{E}_n(\psi_i)\}.$$

It remains to establish a central limit theorem for the centered triangular array. Assumption 4 (iii) gives, for some $\eta > 0$,

$$\sup_n \mathbb{E}_n \|\psi_i\|_2^{2+\eta} < \infty.$$

Jensen's inequality therefore implies

$$\sup_n \mathbb{E}_n \|\psi_i - \mathbb{E}_n(\psi_i)\|_2^{2+\eta} < \infty.$$

For every $\epsilon > 0$,

$$\mathbb{E}_n \left[\|\psi_i - \mathbb{E}_n(\psi_i)\|_2^2 \mathbf{1} \left\{ \|\psi_i - \mathbb{E}_n(\psi_i)\|_2 > \epsilon\sqrt{n} \right\} \right]$$

$$\leq (\epsilon\sqrt{n})^{-\eta} \mathbb{E}_n \|\psi_i - \mathbb{E}_n(\psi_i)\|_2^{2+\eta} \longrightarrow 0.$$

Thus the Lindeberg condition holds for every fixed linear combination of the centered scores.

Since

$$\mathrm{Var}_n(\psi_i) \longrightarrow \Sigma_{\mathrm{miss}},$$

the Lindeberg–Feller theorem and the Cramér–Wold device give

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} \{\psi_i - \mathbb{E}_n(\psi_i)\} \xrightarrow{\mathrm{d}} \mathcal{N}(0, \Sigma_{\mathrm{miss}}).$$

This conclusion remains valid when Σ_{miss} is singular: any scalar projection with zero limiting variance converges to zero by Chebyshev's inequality.

Combining this result with (C.34) yields

$$\sqrt{n}\, \widehat{g}(\beta) \xrightarrow{\mathrm{d}} \mathcal{N}(\tau, \Sigma_{\mathrm{miss}}).$$

Finally, (C.32), (C.33), and Slutsky's theorem imply

$$\sqrt{n}\{\widehat{\beta}(M) - \beta\} \xrightarrow{\mathrm{d}} \mathcal{N}\left(K_M\tau,\, K_M\Sigma_{\mathrm{miss}}K_M^\top\right) = \mathcal{N}\{K_M\tau, V_{\mathrm{miss}}(M)\}.$$

By Assumption 4 (iii), $V_{\mathrm{miss}}(M) \in \mathbb{S}^p_{++}$, so the limiting distribution is nondegenerate.

The same argument applies to either single-source system whenever the corresponding block identifies the full coefficient. Its limiting mean is $Q_{\mathrm{IV}}^{-1}\tau_z$ or $Q_{\mathrm{SV}}^{-1}\tau_w$, respectively, and its limiting covariance is

$$Q_{\mathrm{IV}}^{-1}\Sigma_{\mathrm{IV,miss}}Q_{\mathrm{IV}}^{-T} \quad \text{or} \quad Q_{\mathrm{SV}}^{-1}\Sigma_{\mathrm{SV,miss}}Q_{\mathrm{SV}}^{-T}.$$

These Gaussian limits may be degenerate. Nondegenerate single-source inference additionally requires the corresponding limiting covariance to be positive definite.

□

C.6 Proof of Lemma 6

Proof of Lemma 6. Fix K satisfying $KJ = I_p$. The first-order risk is $\mathrm{AMSE}_\tau(K) = \|K\tau\|_2^2 + \mathrm{tr}(K\Sigma_{\mathrm{miss}}K^\top)$. Because the limiting covariance is common to the local directions under consideration,

$$\begin{aligned} R(K) &= \sup_{\|u\|_2 \le 1} \|KDu\|_2^2 + \mathrm{tr}(K\Sigma_{\mathrm{miss}}K^\top) \\ &= \|KD\|_{\mathrm{op}}^2 + \mathrm{tr}(K\Sigma_{\mathrm{miss}}K^\top). \end{aligned}$$

The last equality follows from the definition of the operator norm and remains valid when D is singular.

For an attainable direction, suppose the estimator's root-n error converges to $\mathcal{N}(K\tau, K\Sigma_{\text{miss}}K^\top)$. Uniform integrability of its squared root-n error implies convergence of the corresponding expectations. Thus $\lim_n n\mathbb{E}_\tau\|\widehat{\beta}-\beta\|_2^2 = \|K\tau\|_2^2 + \operatorname{tr}(K\Sigma_{\text{miss}}K^\top)$. Thus, under the stated uniform-integrability condition, the first-order risk equals the limiting scaled mean squared error.

The supremum is taken after defining the pointwise asymptotic risk. No interchange of a supremum and a limit is used. If some directions are unattainable, taking the supremum over the larger set supplies the stated conservative envelope. □

C.7 Proof of Proposition 7

Proof. We first establish consistency of the score covariance estimator. By Theorem 5 and consistency of the sample design moments,

$$\widehat{\beta}(M_0) - \beta = o_p(1), \qquad \widehat{J} - J = o_p(1). \tag{C.35}$$

The estimated projection coefficients are therefore consistent. The sample disturbance cross moments also converge to their row-specific population counterparts, and replacing the disturbances by the preliminary residuals preserves this convergence. For example,

$$\begin{aligned}&\frac{1}{n}Z^\top\{Y - X\widehat{\beta}(M_0)\} - \mathbb{E}_n(z_i\varsigma_i)\\ &\quad= \left\{\frac{1}{n}Z^\top\varsigma - \mathbb{E}_n(z_i\varsigma_i)\right\} - \frac{1}{n}Z^\top X\{\widehat{\beta}(M_0) - \beta\} = o_p(1).\end{aligned}$$

The same argument applies to the cross moments involving x_i and w_i. Moreover, Cauchy–Schwarz gives

$$\sup_n \|\mathbb{E}_n(z_i\varsigma_i)\|_2 \le \{\mathbb{E}\|z_i\|_2^2\}^{1/2}\{\sup_n \mathbb{E}_n(\varsigma_i^2)\}^{1/2} < \infty,$$

with analogous bounds for x_i and w_i.

Expanding the score formulas in (3.4), each entry of $\widehat{\psi}_i - \psi_i$ can be expressed as a finite sum of entries of

$$x_i x_i^\top, \quad x_i z_i^\top, \quad x_i w_i^\top, \quad z_i z_i^\top, \quad w_i w_i^\top, \quad x_i \varsigma_i, \quad z_i \varsigma_i, \quad w_i \varsigma_i,$$

multiplied by coefficient differences of order $o_p(1)$. The moment conditions imply

$$\frac{1}{n}\sum_{i=1}^{n}\big\{\|x_i x_i^\top\|_{\mathrm{op}}^2 + \|x_i z_i^\top\|_{\mathrm{op}}^2 + \|x_i w_i^\top\|_{\mathrm{op}}^2 + \|z_i z_i^\top\|_{\mathrm{op}}^2 + \|w_i w_i^\top\|_{\mathrm{op}}^2$$

$$+ \|x_i \varsigma_i\|_2^2 + \|z_i \varsigma_i\|_2^2 + \|w_i \varsigma_i\|_2^2\big\} = O_p(1).$$

For the last two terms, the required bounds follow from the decompositions

$$z_i \varsigma_i = z_i\{\varsigma_i - z_i^\top \Sigma_z^{-1}\mathbb{E}_n(z_i\varsigma_i)\} + z_i z_i^\top \Sigma_z^{-1}\mathbb{E}_n(z_i\varsigma_i),$$

$$w_i \varsigma_i = w_i\{\varsigma_i - w_i^\top \Sigma_w^{-1}\mathbb{E}_n(w_i\varsigma_i)\} + w_i w_i^\top \Sigma_w^{-1}\mathbb{E}_n(w_i\varsigma_i),$$

together with the preceding uniform bounds. Since the dimensions are fixed, it follows that

$$\frac{1}{n}\sum_{i=1}^{n}\|\widehat{\psi}_i - \psi_i\|_2^2 = o_p(1). \tag{C.36}$$

The uniform $2+\eta$ score-moment bound implies that each entry of $\{\psi_i - \mathbb{E}_n(\psi_i)\}\{\psi_i - \mathbb{E}_n(\psi_i)\}^\top$ has a uniformly bounded $1+\eta/2$ moment. Independence within each row and the triangular-array weak law of large numbers therefore yield

$$\frac{1}{n}\sum_{i=1}^{n}\{\psi_i - \mathbb{E}_n(\psi_i)\}\{\psi_i - \mathbb{E}_n(\psi_i)\}^\top = \mathrm{Var}_n(\psi_i) + o_p(1)$$

$$= \Sigma_{\mathrm{miss}} + o_p(1).$$

Also, $\frac{1}{n}\sum_{i=1}^{n}\psi_i - \mathbb{E}_n(\psi_i) = O_p(n^{-1/2})$, so replacing population centering by sample centering changes the covariance matrix by $O_p(n^{-1})$. By (C.36) and Cauchy–Schwarz,

$$\frac{1}{n}\sum_{i=1}^{n}\|\widehat{\psi}_i - \psi_i\|_2\|\psi_i\|_2 \le \left\{\frac{1}{n}\sum_{i=1}^{n}\|\widehat{\psi}_i - \psi_i\|_2^2\right\}^{1/2}\left\{\frac{1}{n}\sum_{i=1}^{n}\|\psi_i\|_2^2\right\}^{1/2}$$

$$= o_p(1).$$

These bounds also justify replacing the scores by their plug-in estimates and centering by their sample mean. Consequently,

$$\widehat{\Sigma} - \Sigma_{\mathrm{miss}} = o_p(1). \tag{C.37}$$

We next establish consistency of the risk-optimal sensitivity matrix. Write $R(K) = \|KD\|_{\mathrm{op}}^2 + \mathrm{tr}(K\Sigma_{\mathrm{miss}}K^\top)$ and $\widehat{R}(K) = \|KD\|_{\mathrm{op}}^2 + \mathrm{tr}(K\widehat{\Sigma}K^\top)$. Because $\Sigma_{\mathrm{miss}} \succ 0$ and $\mathrm{rank}(J) = p$, the population objective is coercive and strongly convex, and the constraint set is nonempty. Hence the population problem has a unique minimizer. By (C.35) and (C.37), $\Pr\{\mathrm{rank}(\widehat{J}) = p,\ \widehat{\Sigma} \succ 0\} \longrightarrow 1$. The sample problem likewise has a unique minimizer on this event. We work on this event below and assign the estimator any fixed value on its complement, which does not affect the asserted convergence in probability or distribution.

Since the population and sample problems have different constraint sets, first construct a sample-feasible approximation to K^*:

$$\widetilde{K} = K^* + (I_p - K^*\widehat{J})(\widehat{J}^\top\widehat{J})^{-1}\widehat{J}^\top.$$

Using $K^*J = I_p$ and (C.35), we obtain $\widetilde{K}\widehat{J} = I_p$ and $\widetilde{K} - K^* = o_p(1)$. Sample optimality and (C.37) then give

$$\lambda_{\min}(\widehat{\Sigma})\|\widehat{K}^*\|_{\mathrm{F}}^2 \le \widehat{R}(\widehat{K}^*) \le \widehat{R}(\widetilde{K}) = R(K^*) + o_p(1).$$

Since $\lambda_{\min}(\widehat{\Sigma}) \xrightarrow{\mathrm{p}} \lambda_{\min}(\Sigma_{\mathrm{miss}}) > 0$, it follows that

$$\|\widehat{K}^*\|_{\mathrm{F}} = O_p(1). \tag{C.38}$$

Together with (C.37), this implies

$$\left|\widehat{R}(\widehat{K}^*) - R(\widehat{K}^*)\right| \le \|\widehat{\Sigma} - \Sigma_{\mathrm{miss}}\|_{\mathrm{op}} \|\widehat{K}^*\|_{\mathrm{F}}^2 = o_p(1). \tag{C.39}$$

Next, construct a population-feasible approximation to $\widehat{K}^*$:

$$\overline{K} = \widehat{K}^* + (I_p - \widehat{K}^* J)(J^\top J)^{-1} J^\top.$$

The sample constraint $\widehat{K}^* \widehat{J} = I_p$, together with (C.35) and (C.38), gives $I_p - \widehat{K}^* J = \widehat{K}^*(\widehat{J} - J) = o_p(1)$. Therefore,

$$\overline{K} J = I_p, \qquad \overline{K} - \widehat{K}^* = o_p(1). \tag{C.40}$$

By continuity of R on bounded sets and (C.38), $R(\overline{K}) - R(\widehat{K}^*) = o_p(1)$. Combining this relation with (C.39), population feasibility, and sample optimality yields

$$\begin{aligned} 0 &\le R(\overline{K}) - R(K^*) \\ &= \widehat{R}(\widehat{K}^*) - R(K^*) + o_p(1) \\ &\le \widehat{R}(\widetilde{K}) - R(K^*) + o_p(1) = o_p(1). \end{aligned}$$

Strong convexity and optimality of K^* on $\{K : KJ = I_p\}$ imply

$$\lambda_{\min}(\Sigma_{\mathrm{miss}}) \|\overline{K} - K^*\|_{\mathrm{F}}^2 \le R(\overline{K}) - R(K^*) = o_p(1).$$

Together with (C.40), this establishes

$$\widehat{K}^* - K^* = (\widehat{K}^* - \overline{K}) + (\overline{K} - K^*) = o_p(1). \tag{C.41}$$

Finally, we derive the asymptotic distribution. The sample constraint gives the exact identity

$$\sqrt{n}(\widehat{\beta}_{\text{opt}} - \beta) = \widehat{K}^* \sqrt{n}\, \widehat{g}(\beta). \tag{C.42}$$

The first-order expansion established in the proof of Theorem 5 gives

$$\sqrt{n}\, \widehat{g}(\beta) \xrightarrow{\text{d}} \mathcal{N}(\tau, \Sigma_{\text{miss}}).$$

Applying Slutsky's theorem to (C.41) and (C.42), we obtain

$$\sqrt{n}(\widehat{\beta}_{\text{opt}} - \beta) \xrightarrow{\text{d}} \mathcal{N}\left(K^*\tau,\, K^*\Sigma_{\text{miss}}{K^*}^\top\right).$$

Under exact specification with $D = 0$ and $\Sigma_{\text{miss}} = \Sigma$, the population minimizer and its covariance reduce to

$$K^* = (J^\top \Sigma^{-1} J)^{-1} J^\top \Sigma^{-1}, \qquad K^* \Sigma {K^*}^\top = (J^\top \Sigma^{-1} J)^{-1}.$$

Since $\tau = 0$, the limiting distribution becomes

$$\sqrt{n}(\widehat{\beta}_{\text{opt}} - \beta) \xrightarrow{\text{d}} \mathcal{N}\left(0,\, (J^\top \Sigma^{-1} J)^{-1}\right).$$

□

C.8 Proof of Proposition 8

Proof of Proposition 8. Recall that $R(K) = \|KD\|_{\mathrm{op}}^2 + \mathrm{tr}(K\Sigma_{\mathrm{miss}}K^\top)$, with the constraint $KJ = I_p$. We first establish the risk comparison for a general prespecified budget D, and then derive its exact-specification special case.

The squared operator norm $\|KD\|_{\mathrm{op}}^2$ is convex in K. Since $\Sigma_{\mathrm{miss}} \succ 0$, the quadratic term $\mathrm{tr}(K\Sigma_{\mathrm{miss}}K^\top)$ is strongly convex. Moreover, $R(K) \geq \lambda_{\min}(\Sigma_{\mathrm{miss}})\|K\|_{\mathrm{F}}^2$, so the objective is coercive. The constraint set $\{K : KJ = I_p\}$ is closed and affine. Because $\mathrm{rank}(J) = p$, it is nonempty: $(J^\top J)^{-1}J^\top$ is a feasible left inverse. Continuity and coercivity therefore ensure existence of a minimizer, while strong convexity ensures uniqueness.

The single-source sensitivities are $K^{(\mathrm{IV})} = (Q_{\mathrm{IV}}^{-1}, 0_{p\times p})$ and $K^{(\mathrm{SV})} = (0_{p\times p}, Q_{\mathrm{SV}}^{-1})$. Since $J = (Q_{\mathrm{IV}}^\top, Q_{\mathrm{SV}}^\top)^\top$, we have $K^{(\mathrm{IV})}J = Q_{\mathrm{IV}}^{-1}Q_{\mathrm{IV}} = I_p$ and $K^{(\mathrm{SV})}J = Q_{\mathrm{SV}}^{-1}Q_{\mathrm{SV}} = I_p$. Both are therefore feasible competitors in (4.3). Evaluating all three sensitivities under the same D and Σ_{miss}, optimality gives

$$R(K^*) \leq \min\{R(K^{(\mathrm{IV})}), R(K^{(\mathrm{SV})})\}. \tag{C.43}$$

Each individual inequality requires only that the corresponding single-source block be nonsingular.

Under exact specification, set $D = 0$ and $\Sigma_{\mathrm{miss}} = \Sigma$. The bias term then vanishes for every feasible K, so $R(K) = \mathrm{tr}(K\Sigma K^\top)$. Thus the same risk optimization reduces to minimizing the trace of the asymptotic covariance.

We now verify that its unique minimizer is $K^* = (J^\top\Sigma^{-1}J)^{-1}J^\top\Sigma^{-1}$. Since $\Sigma \succ 0$ and

$\operatorname{rank}(J) = p$, $J^\top \Sigma^{-1} J$ is positive definite, so this expression is well defined. Direct multiplication gives $K^* J = I_p$, $K^* \Sigma = (J^\top \Sigma^{-1} J)^{-1} J^\top$, and $K^* \Sigma K^{*\top} = (J^\top \Sigma^{-1} J)^{-1}$.

To compare this sensitivity with any feasible K, write $L := K - K^*$. This difference satisfies $LJ = 0$ because both sensitivities are left inverses of J. Consequently, $K^* \Sigma L^\top = (J^\top \Sigma^{-1} J)^{-1} (LJ)^\top = 0$, and $L \Sigma K^{*\top} = 0$ by symmetry. The cross terms therefore vanish when expanding the covariance:

$$\begin{aligned} K \Sigma K^\top &= (K^* + L) \Sigma (K^* + L)^\top \\ &= (J^\top \Sigma^{-1} J)^{-1} + L \Sigma L^\top \succeq (J^\top \Sigma^{-1} J)^{-1}. \end{aligned} \tag{C.44}$$

Taking traces in (C.44) gives $R(K) - R(K^*) = \operatorname{tr}(L \Sigma L^\top) = \|L \Sigma^{1/2}\|_{\mathrm{F}}^2 \geq 0$. Since $\Sigma \succ 0$, equality holds only when $L = 0$, which also verifies uniqueness of the stated minimizer.

For the single-source sensitivities, $K^{(\mathrm{IV})} \Sigma K^{(\mathrm{IV})\top} = Q_{\mathrm{IV}}^{-1} \Sigma_{\mathrm{IV}} Q_{\mathrm{IV}}^{-T}$ and $K^{(\mathrm{SV})} \Sigma K^{(\mathrm{SV})\top} = Q_{\mathrm{SV}}^{-1} \Sigma_{\mathrm{SV}} Q_{\mathrm{SV}}^{-T}$. Substituting these expressions into (C.43) yields

$$\operatorname{tr}\{(J^\top \Sigma^{-1} J)^{-1}\} \leq \min \left\{ \operatorname{tr}(Q_{\mathrm{IV}}^{-1} \Sigma_{\mathrm{IV}} Q_{\mathrm{IV}}^{-T}), \operatorname{tr}(Q_{\mathrm{SV}}^{-1} \Sigma_{\mathrm{SV}} Q_{\mathrm{SV}}^{-T}) \right\}.$$

Hence the exact-specification variance comparison is precisely the risk comparison with zero bias budget.

Finally, consider the interpretation in terms of AMSE. For each fixed attainable drift τ, the stated first-order limiting law associated with K is $\mathcal{N}(K\tau, K \Sigma_{\mathrm{miss}} K^\top)$. Its expected squared

Euclidean norm equals $\|K\tau\|_2^2 + \mathrm{tr}(K\Sigma_{\text{miss}}K^\top)$. Using $\mathcal{C}(D) = \{Du : \|u\|_2 \leq 1\}$, we obtain

$$\begin{aligned} &\sup_{\tau \in \mathcal{C}(D)} \left\{ \|K\tau\|_2^2 + \mathrm{tr}(K\Sigma_{\text{miss}}K^\top) \right\} \\ &\quad = \sup_{\|u\|_2 \leq 1} \|KDu\|_2^2 + \mathrm{tr}(K\Sigma_{\text{miss}}K^\top) = R(K). \end{aligned} \tag{C.45}$$

The last equality follows from the definition of the operator norm.

Uniform integrability of squared root-n errors upgrades convergence in distribution to convergence of their expectations for each fixed attainable τ. When every drift in $\mathcal{C}(D)$ is attainable, Lemma 6 and (C.45) therefore identify $R(K)$ with the supremum of pointwise local AMSE. This takes the supremum after the pointwise risk limits; it does not interchange a supremum and a finite-sample limit. If some drifts are unattainable, $R(K)$ remains an envelope over the prespecified ellipsoid.

Under exact specification, Assumption 3 with $M^\star = \Sigma^{-1}$ supplies the centered asymptotic law for the efficient combined estimator. Together with the corresponding single-source limits and uniform integrability, the risks reduce to the covariance traces, establishing the AMSE comparison. The general comparison concerns worst-case risk over $\mathcal{C}(D)$, and does not imply dominance at every fixed τ or finite sample size. □

D. Conditional risk along the restricted WLS path

This appendix considers a one-parameter weighted least-squares (WLS) path combining the IV and SV projection criteria, and characterizes its finite-sample conditional squared-error risk. The path provides an interpretable restriction of the general GMM weighting framework: it

uses block-diagonal weights indexed by a single parameter, whereas unrestricted GMM weights can also account for cross-block dependence through off-diagonal blocks.

We assume that both population information blocks are nonsingular so that both single-source endpoints are identified. This assumption excludes cases such as $q < p$, in which the IV block is rank deficient even though the stacked moment system may remain identified.

For $\lambda \in [0, 1]$, define

$$\widehat{\beta}_\lambda := \operatorname*{argmin}_{\theta \in \mathbb{R}^p} \left\{ (1-\lambda)\|P_Z(Y - X\theta)\|_2^2 + \lambda \|P_W^\perp (Y - X\theta)\|_2^2 \right\}.$$

Whenever the inverse exists, the unique minimizer is

$$\widehat{\beta}_\lambda = \{(1-\lambda)X^\top P_Z X + \lambda X^\top P_W^\perp X\}^{-1}\{(1-\lambda)X^\top P_Z Y + \lambda X^\top P_W^\perp Y\}. \tag{D.1}$$

At the endpoints, this expression recovers $\widehat{\beta}_0 = \widehat{\beta}_{\text{IV}}$ and $\widehat{\beta}_1 = \widehat{\beta}_{\text{SV}}$.

Proposition 9 (Restricted-GMM interpretation)**.** Suppose $n^{-1}X^\top P_Z X \in \mathbb{S}^p_{++}$ and $n^{-1}X^\top P_W^\perp X \in \mathbb{S}^p_{++}$. For $\lambda \in [0, 1]$, define

$$M_\lambda := \operatorname{diag}\left\{ (1-\lambda)(n^{-1}X^\top P_Z X)^{-1}, \lambda (n^{-1}X^\top P_W^\perp X)^{-1} \right\}.$$

Then $\widehat{\beta}_\lambda = \widehat{\beta}(M_\lambda)$. The weight is positive definite for $\lambda \in (0, 1)$ and positive semidefinite at the endpoints.

For fixed λ, sample moment consistency gives $M_\lambda \xrightarrow{\text{p}} \operatorname{diag}\{(1-\lambda)Q_{\text{IV}}^{-1}, \lambda Q_{\text{SV}}^{-1}\}$. The corresponding limiting sensitivity is $\{(1-\lambda)Q_{\text{IV}} + \lambda Q_{\text{SV}}\}^{-1}\big((1-\lambda)I_p, \lambda I_p\big)$. Under the regularity

conditions for the combined GMM estimator, the sample-dependent weight therefore has the same first-order behavior as its deterministic limit. A data-selected λ requires additional conditions on its convergence.

To examine the finite-sample risk along this path, condition on the realized design matrices and write $\mathcal{D} := \sigma(X, Z, W)$. Let $S_Z := X^\top P_Z X$, $S_W := X^\top P_W^\perp X$, and $S_\lambda := (1-\lambda)S_Z + \lambda S_W$. Define $U_Z := X^\top P_Z \varsigma$, $U_W := X^\top P_W^\perp \varsigma$, and $U_\lambda := (1-\lambda)U_Z + \lambda U_W$. Substituting $Y = X\beta + \varsigma$ into (D.1) yields $\widehat{\beta}_\lambda - \beta = S_\lambda^{-1} U_\lambda$. Hence the conditional squared-error risk is

$$\mathcal{R}_{\mathrm{tr}}(\lambda) := \mathbb{E}(\|\widehat{\beta}_\lambda - \beta\|_2^2 \mid \mathcal{D}) = \mathrm{tr}\{S_\lambda^{-1}\mathbb{E}(U_\lambda U_\lambda^\top \mid \mathcal{D})S_\lambda^{-1}\}. \tag{D.2}$$

This expression applies whenever the inverse and conditional second moments exist. Under local misspecification, conditional expectations are evaluated under the row-n distribution.

The analysis does not impose conditional validity of the projected moments. In particular, the risk admits the bias–variance decomposition

$$\mathcal{R}_{\mathrm{tr}}(\lambda) = \|S_\lambda^{-1}\mathbb{E}(U_\lambda \mid \mathcal{D})\|_2^2 + \mathrm{tr}\{S_\lambda^{-1}\,\mathrm{Var}(U_\lambda \mid \mathcal{D})S_\lambda^{-1}\}.$$

Unconditional IV or SV validity does not imply conditional centering and therefore does not justify omitting the first term.

To retain both conditional bias and variance in the risk bounds, define $c_Z := \mathbb{E}(\|U_Z\|_2^2 \mid \mathcal{D})$, $c_{ZW} := \mathbb{E}(U_Z^\top U_W \mid \mathcal{D})$, and $c_W := \mathbb{E}(\|U_W\|_2^2 \mid \mathcal{D})$. These are conditional second moments, rather than conditional variances. Whenever the conditional covariance of ς exists,

$$c_Z = \mathrm{tr}\{X^\top P_Z\,\mathrm{Var}(\varsigma \mid \mathcal{D})P_Z X\} + \|\mathbb{E}(U_Z \mid \mathcal{D})\|_2^2,$$

$$c_{ZW} = \mathrm{tr}\{X^\top P_Z \,\mathrm{Var}(\varsigma \mid \mathcal{D}) P_W^\perp X\} + \mathbb{E}(U_Z \mid \mathcal{D})^\top \mathbb{E}(U_W \mid \mathcal{D}),$$

$$c_W = \mathrm{tr}\{X^\top P_W^\perp \,\mathrm{Var}(\varsigma \mid \mathcal{D}) P_W^\perp X\} + \|\mathbb{E}(U_W \mid \mathcal{D})\|_2^2.$$

The following result bounds the conditional risk using these second moments and the extreme eigenvalues of the information matrices. It also gives a surrogate criterion whose stationary point has a closed-form expression.

Proposition 10 (Conditional risk bounds and a tractable surrogate)**.** Condition on a realized design for which $S_Z, S_W \in \mathbb{S}_{++}^p$ and $c_Z, c_W < \infty$. For every $\lambda \in [0,1]$, the conditional risk in (D.2) satisfies

$$\frac{(1-\lambda)^2 c_Z + 2\lambda(1-\lambda)c_{ZW} + \lambda^2 c_W}{\lambda_{\max}(S_\lambda)^2} \le \mathcal{R}_{\mathrm{tr}}(\lambda) \le \frac{(1-\lambda)^2 c_Z + 2\lambda(1-\lambda)c_{ZW} + \lambda^2 c_W}{\lambda_{\min}(S_\lambda)^2}. \quad \text{(D.3)}$$

Define the surrogate criterion

$$\widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda) := \frac{(1-\lambda)^2 c_Z + 2\lambda(1-\lambda)c_{ZW} + \lambda^2 c_W}{\{(1-\lambda)\lambda_{\min}(S_Z) + \lambda\lambda_{\min}(S_W)\}^2}. \quad \text{(D.4)}$$

Then $\mathcal{R}_{\mathrm{tr}}(\lambda) \le \widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda)$ for all $\lambda \in [0,1]$. If, for a specified constant $C \ge 1$, $\lambda_{\max}(S_Z) \le C\lambda_{\min}(S_Z)$ and $\lambda_{\max}(S_W) \le C\lambda_{\min}(S_W)$, then

$$C^{-2}\widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda) \le \mathcal{R}_{\mathrm{tr}}(\lambda) \le \widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda) \qquad \text{for all } \lambda \in [0,1].$$

For a sequence of designs, a common C gives a comparison that is also uniform in sample size.

If $(c_W - c_{ZW})\lambda_{\min}(S_Z) + (c_Z - c_{ZW})\lambda_{\min}(S_W) \neq 0$, the surrogate has at most one interior stationary point, given by

$$\lambda^\star_{\mathrm{sur}} = \frac{c_Z\lambda_{\min}(S_W) - c_{ZW}\lambda_{\min}(S_Z)}{(c_W - c_{ZW})\lambda_{\min}(S_Z) + (c_Z - c_{ZW})\lambda_{\min}(S_W)}. \quad \text{(D.5)}$$

Its minimum over $[0,1]$ is obtained by comparing its values at the two endpoints and at $\lambda^\star_{\mathrm{sur}}$, when the latter belongs to $(0,1)$. If the displayed denominator is zero, the surrogate is monotone or constant, and an endpoint attains the minimum.

The surrogate risk criterion combines conditional second moments in the numerator with a lower bound on information strength in the denominator. It is an upper bound on the true conditional risk and, when both information matrices have condition numbers bounded by C, any surrogate minimizer has conditional risk at most C^2 times the minimum conditional risk along the WLS path. This guarantee concerns the achieved risk; the surrogate minimizer need not be close to the true risk minimizer.

Moreover, c_Z, c_{ZW}, c_W depend on unknown conditional disturbance moments. The surrogate therefore defines an oracle criterion rather than a feasible tuning rule. In particular, $\lambda^\star_{\mathrm{sur}}$ is only a candidate for minimizing the surrogate and must be compared with the endpoints. These conclusions concern the restricted WLS path and do not characterize the unrestricted GMM optimum.

D.1 Proof of Proposition 9

Proof of Proposition 9. Using $S_Z = X^\top P_Z X$ and $S_W = X^\top P_W^\perp X$, write $M_\lambda = n\,\mathrm{diag}\{(1-\lambda)S_Z^{-1}, \lambda S_W^{-1}\}$ and $\widehat{J} = n^{-1}(S_Z^\top, S_W^\top)^\top$. Since both information matrices are symmetric,

$$\widehat{J}^\top M_\lambda = \big((1-\lambda)I_p, \lambda I_p\big), \qquad \widehat{J}^\top M_\lambda \widehat{J} = \frac{1}{n} S_\lambda.$$

The second matrix is positive definite for every $\lambda \in [0,1]$, because $S_Z, S_W \in \mathbb{S}^p_{++}$.

The GMM first-order condition is therefore

$$(1-\lambda)X^\top P_Z(Y-X\theta)+\lambda X^\top P_W^\perp(Y-X\theta)=0_p.$$

This is also the first-order condition of the WLS criterion. Both criteria are strictly convex in θ, so their unique minimizers coincide: $\widehat{\beta}(M_\lambda)=\widehat{\beta}_\lambda$.

For $\lambda\in(0,1)$, both diagonal blocks of M_λ are positive definite. At either endpoint, one block is zero, making the weight positive semidefinite, while $\widehat{J}^\top M_\lambda \widehat{J}$ remains positive definite. The same calculation thus recovers the IV-only and SV-only estimators. □

D.2 Proof of Proposition 10

Proof of Proposition 10. All expectations and variances in this proof are conditional on $\mathcal{D}=\sigma(X,Z,W)$. Since S_Z, S_W are positive definite, S_λ is positive definite for every $\lambda\in[0,1]$.

Substituting $Y=X\beta+\varsigma$ into (D.1) gives $\widehat{\beta}_\lambda-\beta=S_\lambda^{-1}U_\lambda$. Let $\mu_\lambda:=\mathbb{E}(U_\lambda\mid\mathcal{D})$ and $\Omega_\lambda:=\mathbb{E}(U_\lambda U_\lambda^\top\mid\mathcal{D})$. Because S_λ is $\mathcal{D}$-measurable, the conditional bias is $S_\lambda^{-1}\mu_\lambda$, and

$$\mathcal{R}_{\mathrm{tr}}(\lambda)=\mathrm{tr}(S_\lambda^{-1}\Omega_\lambda S_\lambda^{-1})=\|S_\lambda^{-1}\mu_\lambda\|_2^2+\mathrm{tr}\{S_\lambda^{-1}\,\mathrm{Var}(U_\lambda\mid\mathcal{D})S_\lambda^{-1}\}.$$

The second equality follows from $\Omega_\lambda=\mathrm{Var}(U_\lambda\mid\mathcal{D})+\mu_\lambda\mu_\lambda^\top$. The expressions for c_Z, c_{ZW}, c_W follow by applying the same second-moment decomposition to U_Z and U_W.

To establish the risk bounds, expand $U_\lambda=(1-\lambda)U_Z+\lambda U_W$ and write

$$N(\lambda):=\mathrm{tr}(\Omega_\lambda)=(1-\lambda)^2c_Z+2\lambda(1-\lambda)c_{ZW}+\lambda^2c_W.$$

This quantity is finite and nonnegative because $N(\lambda) = \mathbb{E}(\|U_\lambda\|_2^2 \mid \mathcal{D})$. Finiteness of c_{ZW} follows from conditional Cauchy–Schwarz.

For positive semidefinite matrices A and B, $\lambda_{\min}(B)\,\mathrm{tr}(A) \le \mathrm{tr}(AB) \le \lambda_{\max}(B)\,\mathrm{tr}(A)$. These inequalities follow by conjugating $\lambda_{\min}(B)I \preceq B \preceq \lambda_{\max}(B)I$ by $A^{1/2}$ and taking traces. Applying them to $\mathcal{R}_{\mathrm{tr}}(\lambda) = \mathrm{tr}(\Omega_\lambda S_\lambda^{-2})$ yields

$$\frac{N(\lambda)}{\lambda_{\max}(S_\lambda)^2} \le \mathcal{R}_{\mathrm{tr}}(\lambda) \le \frac{N(\lambda)}{\lambda_{\min}(S_\lambda)^2},$$

which proves (D.3).

For the surrogate comparison, let $a_Z := \lambda_{\min}(S_Z)$, $a_W := \lambda_{\min}(S_W)$, and $d_\lambda := (1-\lambda)a_Z + \lambda a_W$. These quantities describe the lower bound on information strength used in the surrogate. Since $a_Z, a_W > 0$, we have $d_\lambda > 0$ throughout $[0,1]$. The Rayleigh-quotient characterization gives

$$\lambda_{\min}(S_\lambda) \ge d_\lambda, \qquad \lambda_{\max}(S_\lambda) \le (1-\lambda)\lambda_{\max}(S_Z) + \lambda\lambda_{\max}(S_W).$$

Consequently, $\mathcal{R}_{\mathrm{tr}}(\lambda) \le N(\lambda)/d_\lambda^2 = \widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda)$. Under the stated condition-number bounds, $\lambda_{\max}(S_\lambda) \le C d_\lambda$. Combining this inequality with the lower risk bound gives

$$C^{-2}\widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda) \le \mathcal{R}_{\mathrm{tr}}(\lambda) \le \widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda) \qquad \text{for all } \lambda \in [0,1].$$

The comparison is also uniform in sample size when the same C applies throughout a sequence of designs.

It remains to characterize the stationary points of the surrogate. Write

$$N(\lambda) = c_Z + 2(c_{ZW} - c_Z)\lambda + (c_Z - 2c_{ZW} + c_W)\lambda^2.$$

Since $\widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda) = N(\lambda)/d_\lambda^2$ and $d'_\lambda = a_W - a_Z$, differentiation yields

$$\widetilde{\mathcal{R}}'_{\mathrm{tr}}(\lambda) = \frac{N'(\lambda)d_\lambda - 2N(\lambda)(a_W - a_Z)}{d_\lambda^3} = \frac{2\big[\{(c_W - c_{ZW})a_Z + (c_Z - c_{ZW})a_W\}\lambda - (c_Z a_W - c_{ZW}a_Z)\big]}{d_\lambda^3}.$$

The quadratic terms in the numerator cancel. If $(c_W - c_{ZW})a_Z + (c_Z - c_{ZW})a_W \neq 0$, the derivative has at most one zero, given by

$$\lambda^\star_{\mathrm{sur}} = \frac{c_Z a_W - c_{ZW}a_Z}{(c_W - c_{ZW})a_Z + (c_Z - c_{ZW})a_W}.$$

Substituting the definitions of a_Z and a_W gives (D.5).

If instead $(c_W - c_{ZW})a_Z + (c_Z - c_{ZW})a_W = 0$, the derivative equals $-2(c_Z a_W - c_{ZW}a_Z)/d_\lambda^3$. Because $d_\lambda > 0$, the surrogate is strictly monotone when $c_Z a_W - c_{ZW}a_Z \neq 0$, and constant otherwise. In either case, an endpoint attains the minimum.

The surrogate is continuous on the compact interval $[0, 1]$, so it attains a minimum. In the nondegenerate case, every interior minimizer must be the stationary point above. It therefore suffices to compare the two endpoint values and the value at $\lambda^\star_{\mathrm{sur}}$, when this candidate belongs to $(0, 1)$.

Finally, the uniform risk comparison implies

$$\min_{\lambda\in[0,1]} \widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda) \leq C^2 \min_{\lambda\in[0,1]} \mathcal{R}_{\mathrm{tr}}(\lambda).$$

At any surrogate minimizer, the true conditional risk is bounded above by the surrogate minimum. Its conditional risk is therefore at most C^2 times the minimum true conditional risk along the path. □

E. Data preparation and implementation for the real-data analysis

Experimental source and batch-level profiles. We use the K562 essential-scale, day-6 Perturb-seq dataset of Replogle et al. (2022). K562 cells expressing dCas9–KRAB were transduced with a pooled lentiviral library targeting common essential genes defined by the Cancer Dependency Map. Each targeting vector expressed two single-guide RNAs directed at the same gene; non-targeting controls constituted approximately 5% of the library. GFP-positive transduced cells were enriched by fluorescence-activated cell sorting three days after infection. Six days after infection, cells were profiled using 10x Genomics Chromium single-cell RNA sequencing with direct guide capture across 48 lanes/GEM groups. These sequencing groups define the batches in our analysis.

The supplied data contain 309,915 input cells and 8,563 genes. The precomputed pseudobulk matrix groups cells by perturbation and batch and excludes groups containing fewer than five cells. Within each retained group, five cells are sampled with replacement, and their log1p expression values are averaged gene by gene. This produces 22,075 perturbation–batch profiles. The resulting observations are therefore averages of transformed expression from sampled cells, rather than sums of raw counts.

Perturbation and outcome selection. We retain a perturbation if it has at least 30 pseudobulk profiles and both batch-stratified Wilcoxon results and target-gene expression

measurements are available. The retained data comprise 213 perturbations and 48 control profiles, totaling 8,413 profiles. Each perturbation contributes 30–48 profiles, with a median of 39. Combining each perturbation with the same 48 control profiles gives target-specific sample sizes of 78–96, with a median of 87. All target-specific analyses contain 48 batches.

For each retained perturbation, we rank the 8,563 genes by increasing Wilcoxon adjusted p-value, resolving ties by decreasing absolute z-score and then decreasing absolute log fold change. We average each gene's rank across the 213 perturbations. After excluding all retained perturbation targets, the 2,000 genes with the smallest average ranks form the common outcome panel. Ties are resolved by decreasing mean absolute z-score, decreasing mean absolute log fold change, and gene name. Every retained perturbation is analyzed against every outcome, yielding 426,000 fitted models.

Surrogate-gene selection and PCA. We exclude all retained perturbation targets and all genes appearing among the top 20 DE-ranked genes for any retained perturbation. Among the remaining genes, we select the 1,000 with the largest average DE ranks, resolving ties by increasing mean absolute z-score, increasing mean absolute log fold change, and gene name. This rule selects genes with relatively weak evidence of perturbation-associated differential expression; it does not require every selected gene to be nonsignificant under every perturbation.

PCA is fitted once using these genes across all 8,413 retained perturbation and control profiles. Each gene is centered and divided by its standard deviation across profiles; features

with standard deviation at most 10^{-8} are removed. We retain the first 10 principal component scores as surrogate controls. Together, they explain approximately 3.53% of the standardized expression variance. For each target-specific model, we subset these common scores to the relevant perturbation and control profiles.

Estimation and batch-clustered inference. Within each target-specific sample, let z_i^c, x_i^c, and y_i^c denote centered perturbation indicators and expression values. Let $\widetilde{x}_i$ and $\widetilde{y}_i$ denote residuals from separate least-squares regressions of x_i and y_i on an intercept and the 10 principal component scores. The implemented moment vector is

$$g_i(\beta) = \begin{pmatrix} z_i^c(y_i^c - \beta x_i^c) \\ \\ \widetilde{x}_i(\widetilde{y}_i - \beta \widetilde{x}_i) \end{pmatrix}.$$

IV and SV use the first and second moment conditions, respectively. Two-step GMM uses identity weighting in the first step and the generalized inverse of the batch-clustered moment covariance evaluated at the first-step estimate in the second step. Standard errors for all estimators are clustered by batch.

For local misspecification-adjusted inference, we evaluate $\widehat{g}_i = g_i(\widehat{\beta}_{\mathrm{GMM}})$ and compute

$$\widehat{\Omega} = \frac{G}{G-1}\frac{1}{n}\sum_{b=1}^{G}\left\{\sum_{i\in\mathcal{I}_b}(\widehat{g}_i - \bar{g})\right\}\left\{\sum_{i\in\mathcal{I}_b}(\widehat{g}_i - \bar{g})\right\}^{\top},$$

where $G = 48$, $\mathcal{I}_b$ indexes profiles in batch b, and $\bar{g} = n^{-1}\sum_i \widehat{g}_i$. We set $r_j = 0.5\sqrt{\widehat{\Omega}_{jj}}$ and use the same ellipsoidal deviation set for IV, SV, and GMM. The adjusted inference accounts for the worst-case local bias over this set while retaining each method's point estimate.

Multiple testing and biological enrichment. For each perturbation, we apply Benjamini–Hochberg adjustment across the 2,000 outcomes separately for each estimator and each inference rule. For the Wilcoxon comparison, we apply the same Benjamini–Hochberg adjustment to the raw p-values for the 2,000 outcomes within each perturbation. GO biological-process enrichment uses the tested outcome panel as its background: 1,790 of the 2,000 genes have direct GO:BP annotations, and 4,795 terms are tested for each perturbation-specific discovery set.